t3-w06 — Construction — Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

Run `c0c3-deepseek-native` · conditions C0, C1, C2, C3 · the pack shown is draw 3 of each.

HER LESSON PLAN

Construction — Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

Objectives

1. construct parallel lines
2. construct perpendicular lines
3. bisect a given line segment
4. construct angle 90°
5. construct angle 60°

Step 1 — intersecting lines

Teacher draws intersecting lines and ask students to measure some angles, while the students measure some angles.

- **teacher** draws intersecting lines and asks students to measure some angles
- **student** measure some angles

Step 2 — parallel lines

Teacher draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws, while the students follow and replicate it in their notes.

- **teacher** draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws
- **student** construct parallel lines, following the teacher's board construction and replicating it in their notes

Step 3 — perpendicular lines

Teacher draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws, while the students follow and replicate it in their notes.

- **teacher** draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws
- **student** follow the teacher's board construction and replicate it in their notes

Step 4 — construction of line segments

Teacher draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws, while the students follow and replicate it in their notes.

- **teacher** draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws
- **student** construct line segment, following the teacher's board construction and replicating it in their notes

Step 5 — angle 90°

Teacher constructs angle 90° on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs, while the students follow and replicate it in their notes.

- **teacher** constructs angle 90 degrees on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs
- **student** construct angle 90 degrees, following the teacher's board construction and replicating it in their notes

Step 6 — angle 60°

Teacher constructs angle 60° on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs, while the students follow and replicate it in their notes.

- **teacher** constructs angle 60 degrees on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs
- **student** construct angle 60 degrees, following the teacher's board construction and replicating it in their notes

HER LESSON NOTE

Verbatim, as transcribed. **C2 and C3 receive this; C0 and C1 never do** — which is the whole of what separates them.

Step 2 — parallel lines

```
**Examples:** construct a line through Point P so that it is parallel to AB.

(a) Place a set square so that one edge is accuratly along AB

**fig-01.** Line $AB$, with a point $P$ marked above it. A set square is drawn below $AB$ with
one edge along the line: the right angle is at the left end of that edge, one leg runs down from
it, and the hypotenuse closes back to a point on $AB$ to the right.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \node[left] at (0,0) {$A$};
  \node[above] at (7,0) {$B$};
  \fill (3.4,2.8) circle (1.6pt);
  \node[above] at (3.4,2.8) {$P$};
  \draw (2.7,0) -- (2.7,-1.1) -- (5.2,0);
  \draw (2.7,-0.26) -- (2.96,-0.26) -- (2.96,0);
\end{tikzpicture}
\end{center}

(b) Place a ruler along one of the other edge of the set square. Use the left hand edge if you
are right handed.

**fig-02.** The same set square on $AB$, with the ruler now drawn as a long strip lying along the
set square's downward edge. The set square's vertical edge and the ruler's right edge are
**layered** — only one line is visible on the page. The right angle marked at the corner is the
set square's own, between the edge lying along $AB$ and the edge rising from it.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above] at (8,0) {$B$};
  \draw[rounded corners=3pt] (1.29,3.50) rectangle (1.72,-4.12);
  % The set square's vertical edge lies ALONG the ruler's right edge (x=1.72) — in the
  % source the two are layered, so only one line is visible. Drawn collinear, not offset.
  \draw (1.72,0) -- (1.72,-1.9) -- (5.4,0);
  \draw (1.72,-0.26) -- (1.98,-0.26) -- (1.98,0);
\end{tikzpicture}
\end{center}

**fig-03.** The same arrangement drawn a second time alongside the first — the set square's edge
again **layered** on the ruler's right edge, with no gap between them — and with the ruler
leaning
— running from upper left to lower right across $AB$ instead of standing vertical. Only $A$ is
labelled. The slanted ruler is the point of this panel — it shows that the ruler's inclination
does not affect the construction.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
```

```
  \draw (-0.13,3.56) -- (3.426,-3.179) -- (3.046,-3.379) -- (-0.51,3.36) -- cycle;
  % The set square's edge is LAYERED on the ruler's right edge, as in fig-02 — collinear
  % with it and starting exactly at its crossing with AB (x=1.749), no gap.
  \draw (1.749,0) -- (2.377,-1.192) -- (5.11,0);
\end{tikzpicture}
\end{center}

(c) Hold the ruler firmly, slide the set square along the ruler towards P. Stop when the edge
that
was on AB reaches P. Draw a line along this edge of the set square through P.

**fig-04.** The vertical ruler held fixed, with the set square drawn twice — once with its edge
on
$AB$ and once higher up, its edge on a second line parallel to $AB$ — and an arrow between the
two
positions pointing up along the ruler.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above] at (8,0) {$B$};
  \draw (0.71,2.40) -- (8,2.40);
  \draw[rounded corners=3pt] (0.28,3.65) rectangle (0.71,-3.97);
  % ONE set square drawn in two positions, translated straight up the ruler's right edge
  % (x=0.71). Both are congruent — leg 1.21, width 2.68 — so the two hypotenuses are
  % exactly parallel, which is the visible signature of a slide.
  \draw (0.71,0) -- (0.71,-1.21) -- (3.39,0);
  \draw (0.71,-0.26) -- (0.97,-0.26) -- (0.97,0);
  \draw (0.71,2.40) -- (0.71,1.19) -- (3.39,2.40);
  \draw (0.71,2.14) -- (0.97,2.14) -- (0.97,2.40);
  \draw[->] (2.44,0.1) -- (2.44,1.37);
\end{tikzpicture}
\end{center}

**fig-05.** The same slide drawn a second time alongside it, with the leaning ruler. Both set
square positions are **layered** on the ruler's right edge, as in fig-03. Here the upper
line carries the point $P$, and the set square's upper position has its edge running through $P$.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \draw (0.15,2.63) -- (8,2.63);
  \fill (3.24,2.63) circle (1.6pt);
  \node[above] at (3.24,2.63) {$P$};
  \draw (-0.33,3.58) -- (3.14,-3.20) -- (2.755,-3.392) -- (-0.715,3.388) -- cycle;
  % ONE set square in two positions, translated ALONG the ruler's right edge — layered on
  % it, each starting exactly at its crossing (y=2.63 at x=0.156; AB at x=1.502). Both are
  % congruent (leg 1.26, width 3.084), so the edge running from the ruler to AB and the one
  % running from the ruler to P are exactly parallel. The upper edge meets P by construction.
  \draw (0.156,2.63) -- (0.730,1.508) -- (3.240,2.63);
  \draw (1.502,0) -- (2.076,-1.122) -- (4.586,0);
  \draw[->] (3.21,0.1) -- (2.467,1.551);
\end{tikzpicture}
\end{center}
```

Step 3 — perpendicular lines

```
Two lines are said to be perpendicular to each other when they meet at a right angle i.e
$90^\circ$

**fig-06.** Line $AB$ with a second line $XY$ crossing it, $X$ above and $Y$ below, and a
right-angle mark in the upper right quadrant at the crossing.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (-0.5,0) -- (7.5,0);
  \draw (0,0.25) -- (0,-0.25);
  \draw (5,0.25) -- (5,-0.25);
  \node[below] at (0,-0.3) {$A$};
  \node[below] at (5,-0.3) {$B$};
  \draw (2.5,2.6) -- (2.5,-2.6);
  \node[above] at (2.5,2.6) {$X$};
  \node[below] at (2.5,-2.6) {$Y$};
  \draw (2.82,0) -- (2.82,0.32) -- (2.5,0.32);
\end{tikzpicture}
\end{center}

1. Draw a line AB mark a point Q on the line.

**fig-07.** A plain line with a stroke at each end, labelled $A$ on the left and $B$ on the
right.
The point $Q$ named in the instruction is not marked on the drawing.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below] at (0,-0.28) {$A$};
  \node[below] at (7,-0.28) {$B$};
\end{tikzpicture}
\end{center}

2. Place the edge of the set square on line AB. Let the edge of the ruler meet the hypotenuse of
   the set square as shown.

**fig-08.** Line $AB$ with the set square standing on it as a narrow right-angled triangle: a
short base along $AB$, a near-vertical edge rising from the right end of that base, and the
hypotenuse running up from the left.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below] at (0.41,-0.05) {$A$};
  \node[above] at (3.93,0.05) {$B$};
  \draw (1.54,0) -- (2.20,1.0) -- (2.25,0);
\end{tikzpicture}
\end{center}

3. Pressing firmly on the ruler, slide the set square down along the hypotenuse side until the
   third edges touches point Q. Draw a line XY

**fig-09.** Line $AB$ with $XY$ now drawn across it, $X$ above and $Y$ below, and a right-angle
mark in the upper right quadrant. The foot of $XY$ is not labelled $Q$.
```

```
\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below] at (0.15,-0.1) {$A$};
  \node[above] at (4.52,0.1) {$B$};
  \draw (2.1,1.85) -- (2.15,-0.8);
  \node[above] at (2.1,1.85) {$X$};
  \node[below] at (2.15,-0.8) {$Y$};
  \draw (2.44,0) -- (2.44,0.32) -- (2.12,0.32);
\end{tikzpicture}
\end{center}
```

Step 4 — construction of line segments

```
**Procedures:**

1. Open a pair of compasses so that the radius is about $\frac{3}{4}$ of the length of AB.
2. To bisect the line segment AB means to divide it into two equal halves

**fig-10.** The line segment $AB$, a stroke at each end, $A$ on the left and $B$ on the right.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (7,-0.1) {$B$};
\end{tikzpicture}
\end{center}

3. Place the sharp point of the compass on A, Draw two arcs, one above and the other below the
   midle of AB.

**fig-11.** The segment $AB$ with arc pairs crossing above and below it, the upper crossing
labelled $P$ and the lower one $Q$. Each crossing is drawn as a small X where an arc from $A$
meets
an arc from $B$.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (6,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (6,0.22) -- (6,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (6,-0.1) {$B$};
  \draw (0,0) ++(37:4.5) arc (37:59:4.5);
  \draw (6,0) ++(121:4.5) arc (121:143:4.5);
  \draw (0,0) ++(-59:4.5) arc (-59:-37:4.5);
  \draw (6,0) ++(-143:4.5) arc (-143:-121:4.5);
  \node[above] at (3,3.5) {$P$};
  \node[below] at (3,-3.5) {$Q$};
\end{tikzpicture}
\end{center}

4. With the same radius, and place the sharp point of the compasses on B Draw two arc so that
they
   cut the first arc at P and Q
```

```
**fig-12.** The same arcs with $P$ joined to $Q$, the joining line crossing $AB$ at its middle
and
carrying a right-angle mark in the upper right quadrant.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (6,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (6,0.22) -- (6,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (6,-0.1) {$B$};
  \draw (0,0) ++(37:4.5) arc (37:59:4.5);
  \draw (6,0) ++(121:4.5) arc (121:143:4.5);
  \draw (0,0) ++(-59:4.5) arc (-59:-37:4.5);
  \draw (6,0) ++(-143:4.5) arc (-143:-121:4.5);
  \draw (3,3.35) -- (3,-3.35);
  \node[above] at (3,3.35) {$P$};
  \node[below] at (3,-3.35) {$Q$};
  \draw (3.32,0) -- (3.32,0.32) -- (3,0.32);
\end{tikzpicture}
\end{center}
```

Step 5 — angle 90°

```
**Procedure:**

1. Draw a line segment $AB$ and mark a point $O$ on it where the right angle is to stand.
2. With centre $O$ and any convenient radius, draw an arc to cut $AB$ at $X$ and $Y$.
3. With centre $X$ and a radius greater than $XO$, draw an arc above $AB$.
4. With centre $Y$ and the same radius as in (3), draw an arc to cut the first arc at $Z$.
5. Join $OZ$. Then $\angle ZOA = \angle ZOB = 90^\circ$.

**fig-13.** The line $AB$ with a stroke at each end, the two arc cuts $X$ and $Y$ marked either
side of the middle, and the perpendicular $OZ$ erected from $O$ between them, with a right-angle
mark at its foot and the two arcs crossing at $Z$ near its top.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (7,-0.1) {$B$};
  \draw (0.29,0.22) -- (0.29,-0.22);
  \draw (0.84,0.34) -- (0.84,-0.34);
  \node[below] at (0.84,-0.38) {$X$};
  \draw (6.13,0.22) -- (6.13,-0.22);
  \node[below] at (6.13,-0.26) {$Y$};
  \draw (6.60,0.34) -- (6.60,-0.34);
  \draw (3.8,0) -- (3.8,3.2);
  \node[below] at (3.8,-0.12) {$O$};
  \draw (4.12,0) -- (4.12,0.32) -- (3.8,0.32);
  \draw (2.9,2.55) -- (4.7,3.05);
  \draw (2.9,3.05) -- (4.7,2.55);
  \node[above right] at (3.8,2.82) {$Z$};
\end{tikzpicture}
\end{center}
```

Step 6 — angle 60°

```
**Procedure:**

1. Draw a line segment BC with centre B at any convinent radius. Draw an arc to cut BC at X
2. with centre X and the same radius as in (i) draw an arc to cut the first arc at A
3. Join AB.

**fig-14.** The line $BC$ with $B$ at the left, a stroke marking $X$ part way along, a long arc
swung from $B$ through $X$ and up to the left, a shorter arc swung from $X$ cutting it, and a
straight line from $B$ to that crossing. The crossing itself is not labelled.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \node[below left] at (0,-0.05) {$B$};
  \node[above right] at (7,0) {$C$};
  \draw (5.17,0.22) -- (5.17,-0.22);
  \node[below] at (5.17,-0.25) {$X$};
  \draw (0,0) ++(5:5.17) arc (5:74:5.17);
  \draw (5.17,0) ++(103:5.17) arc (103:140:5.17);
  \draw (0,0) -- (2.59,4.48);
\end{tikzpicture}
\end{center}
```

C0 `The curriculum objective alone. No lesson plan, no note.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Construction

`subtopic`

Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

`term`

Third Term

`week`

6

`lesson_objectives`

• construct parallel lines
• construct perpendicular lines
• bisect a given line segment
• construct angle 90°
• construct angle 60°

Objective 1 — construct parallel lines

`objective_text`

construct parallel lines

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 2 — construct perpendicular lines

`objective_text`

```
construct perpendicular lines
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 3 — bisect a given line segment

`objective_text`

```
bisect a given line segment
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 4 — construct angle 90°

`objective_text`

```
construct angle $90^\circ$
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 5 — construct angle 60°

`objective_text`

```
construct angle $60^\circ$
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Drawing Parallel Lines with a Ruler and Set Square

> EXPLANATION
> Parallel lines are lines that run side by side and never meet, no matter how far you extend them. They stay the same distance apart all along their length. When you look at railway tracks or the lines on a ruled page, you are looking at parallel lines.
>
> You can draw parallel lines with just a ruler and a set square. The set square is the plastic triangle you use in class. The trick is to let the ruler stay still and to slide the set square along it.
>
> Here is the way to do it:
>
> 1. Place your ruler on the paper where you want it.
> 2. Put one edge of the set square firmly against the ruler, with the set square resting on the paper.
> 3. Hold the ruler very still. Slide the set square along the ruler to the spot where you want the new line.
> 4. Draw along the free edge of the set square. That line is parallel to the first line.
>
> Because the set square is a rigid triangle, when you slide it along the ruler its edge always makes the same angle with the ruler. So every line you draw along that edge is parallel to every other line you draw the same way.
>
> You can draw several parallel lines this way by sliding the set square to different positions along the same ruler, drawing a line each time. All of those lines are parallel to one another.

> WORKED_EXAMPLE
> Example: Draw a line and then construct a second line parallel to it, 3 cm away.
>
> Step 1 – Draw your first line. Place the ruler on your paper and draw a straight line along its edge with a sharp pencil. Label the line l.
>
> Step 2 – Put the set square down. Hold the ruler still. Place one edge of the set square against the ruler, as shown in the routine above.
>
> Step 3 – Slide the set square. Keep the ruler firmly in place and slide the set square along the ruler until its free edge is 3 cm from your line l.
>
> Step 4 – Draw the second line. With the set square held still, draw along its free edge. This new line runs side by side with l and is the same 3 cm away from it all along its length, so the two lines are parallel.
>
> You can check your work by measuring the distance between the two lines at a few different points with your ruler. If the distance is the same everywhere, your lines are parallel.

1. Draw a horizontal line and label it a. Then construct a second line parallel to it, 4 cm above it. Which tool do you hold still while you slide the other tool?

answer: You hold the ruler still and slide the set square along it.

2. Using the same ruler and set square method, draw three parallel lines on your page. How can you check that the three lines really are parallel?

answer: Measure the distance between the lines with your ruler at several points. If the gaps between neighbouring lines stay the same all the way along, they are parallel.

Building a Right Angle: Constructing Perpendicular Lines

EXPLANATION

Two lines are called perpendicular when they meet at a right angle — that is, the angle between them is exactly 90°. You will often hear this described as "at right angles". Think of the corner of a book or a page: the two edges that meet at the corner are perpendicular to each other.

A small square symbol drawn at the meeting point (∟) is the sign that the angle there is 90°.

In this section you will learn to draw a line that is perpendicular to a given line, using only a ruler and a pair of compasses. The compass lets you mark off equal distances, and it is those equal distances that guarantee the angle comes out exactly right — you do not need to guess or measure with a protractor.

EXPLANATION

How to draw a perpendicular through a point on the line

Suppose you have a line and a point on it, and you want to draw a line through that point that meets the original line at a right angle.

Step 1. Place the point of your compass on the given point and draw an arc that crosses the line on both sides of that point. This marks two points on the line that are equally far from your point.

Step 2. Widen the compass a little. Put the point on one of those two marks and draw an arc above (or below) the line. Keep the compass at the same width.

Step 3. Without changing the width, put the point on the other mark and draw another arc so that the two arcs cross. The crossing point of the two arcs is equally far from both marks.

Step 4. Use your ruler to join the given point to the crossing point of the arcs. That joined line is perpendicular to the original line.

Why does this work? Because your compass guarantees that the two marked points on the line are equally far from the given point, and that the arc crossing point is equally far from both marks. A point that is equally far from the ends of a line segment always lies on the line that cuts that segment in half at right angles — the perpendicular bisector — so your final line is at 90° to the original.

WORKED_EXAMPLE

Worked example 1: a perpendicular through a point on the line

Construct a line perpendicular to AB through the point P, which lies on AB.

Step 1. Place the compass point on P. Draw an arc that cuts AB on both sides of P, giving two marks (call them C and D). Because one arc drew both marks, PC = PD.

Step 2. Widen the compass. Put the point on C and draw an arc above the line.

Step 3. Keep the same width. Put the point on D and draw an arc that crosses the first one. Call the crossing point E.

Step 4. Join P to E with a straight line. The line PE is perpendicular to AB, so the angle between AB and PE is 90°.

WORKED_EXAMPLE

Worked example 2: a perpendicular through a point off the line

Now suppose the given point Q is not on the line, but sits above it, and you want the perpendicular from Q down to the line.

Step 1. Put the compass point on Q. Open it wide enough to reach the line, and draw an arc that crosses the line at two places (call them F and G).

Step 2. Now F and G are on the line. Put the compass point on F and draw an arc below the line.

Step 3. Keep the same width, put the point on G, and draw an arc that crosses the previous one. Call the crossing point H.

> Step 4. Join Q to H with a ruler. The line QH crosses the original line at a right angle, so QH is perpendicular to it, and H lies on the perpendicular from Q.

1. You have a straight line XY drawn on your page, with a point Z lying on the line. Using only a ruler and compass, describe the first two steps you would take to construct a line through Z that is perpendicular to XY.

answer: Step 1: Place the compass point on Z and draw an arc that cuts the line XY on both sides of Z, making two marks on the line. Step 2: Widen the compass a little, place the point on one of those marks, and draw an arc above the line.

2. In the construction for a perpendicular through a point on a line, you join the given point to the crossing point of two arcs. Why does that crossing point guarantee the joined line is at exactly 90° to the original line?

answer: Because the crossing point is equally far from the two marks you made on the line, and a point that is equally far from the two ends of a segment lies on the line that bisects the segment at right angles. So the line joining the given point to that crossing point is perpendicular to the original line.

3. A point Q is drawn above a line, not on the line. To construct the perpendicular from Q down to the line, you first draw an arc from Q that crosses the line at two places. What do you do next, and why must the compass width stay the same for the following two arcs?

answer: Next you place the compass point on one of the two crossing places and draw an arc below the line, then without changing the width place the point on the other crossing place and draw an arc that crosses the first. Keeping the width the same makes both arcs the same size, so their crossing point is equally far from both marks on the line, which places it on the perpendicular.

Splitting a Line Segment Exactly in Half

EXPLANATION
To bisect a line segment means to cut it into two equal halves. In construction, we do this without measuring the length with a ruler — we use a compass and a straight edge (your ruler, used only to draw straight lines). When we bisect a segment AB, we find the point M exactly in the middle, so that AM = MB. At the same time we draw a line through M at right angles (90°) to AB. That line is called the perpendicular bisector of AB.

WORKED_EXAMPLE
Worked example: Bisecting segment AB

Draw a segment AB about 8 cm long. Follow these steps.

Step 1 — Set your compass
Open your compass to a width that is more than half the length of AB. A good choice is about three quarters of the way across. Keep this same opening for the rest of the construction.

Step 2 — Draw arcs from A
Put the compass point on A and draw an arc that passes above and below the segment.

Step 3 — Draw arcs from B
Keep the compass at the same opening. Put the compass point on B and draw another arc. The two arcs cross at two points — one above AB and one below it. Call them C and D.

Step 4 — Join C and D
Use your straight edge to draw a line through C and D. This line crosses AB at a point M.

Step 5 — Check your work
M is exactly the middle of AB, so AM = MB. The line CD also meets AB at a right angle (90°). The line CD is the perpendicular bisector of AB.

EXPLANATION
Why must the compass opening be more than half of AB? If the opening were less than half, the two arcs would not meet at all, and you could not find points C and D. If the opening were exactly half, the arcs would meet at only one point, right in the middle — but then you could not draw a line through two points, and you would have no perpendicular bisector. That is why the rule is: open the compass wider than half of AB, and do not change the opening between drawing the arcs from A and from B.

You can check that M is the middle by placing your compass point on M, opening to A, and turning the compass — the point should land exactly on B.

WORKED_EXAMPLE
Worked example: Bisecting a shorter segment

Now bisect a segment PQ that is only about 5 cm long.

Step 1 — Open your compass wider than half of PQ (for example, about 3.5 cm).
Step 2 — With the compass on P, draw arcs above and below the segment.
Step 3 — With the same opening and the compass on Q, draw arcs that cross the first two. The crossing points are R and S.
Step 4 — Draw a straight line through R and S. It cuts PQ at N.
Step 5 — N is the midpoint: PN = NQ, and the line RS is at right angles to PQ.

Notice the method is exactly the same whether the segment is long or short — you only change the compass opening so it is always more than half the segment's length.

1. You are given a segment XY and asked to bisect it. You open your compass to a width that is less than half of XY and draw arcs from X and from Y. What will happen, and what should you do instead?

answer: The two arcs will not meet, so you will not be able to find the two crossing points. You should open the compass wider, so that it is more than half of XY, and keep that same opening while drawing arcs from both X and Y.

2. You are bisecting segment AB and you have drawn your arcs and joined the two crossing points C and D with a straight line. The line CD crosses AB at a point M. What can you say about the lengths AM and MB, and what special name do we give to the line CD?

answer: AM = MB (M is the midpoint of AB, so the segment is cut into two equal halves), and CD is called the perpendicular bisector of AB because it also meets AB at a right angle (90°).

3. Describe, in the correct order, the steps you would take to bisect a segment AB using a compass and a straight edge. Include what you must remember about the compass opening.

answer: 1. Open the compass so it is more than half of AB and keep that opening. 2. With the point on A, draw arcs above and below AB. 3. With the same opening and the point on B, draw arcs that cross the first two; call the crossings C and D. 4. Draw a straight line through C and D. 5. Where it cuts AB is the midpoint M, with AM = MB, and CD is the perpendicular bisector.

Constructing a Right Angle (90°)

EXPLANATION

A right angle measures exactly 90°. You can recognise it as the corner of a book, the corner of a page, or the angle made when two straight lines stand exactly perpendicular to each other.

In construction, we draw a 90° angle using only two simple tools: a ruler and a pair of compasses. You never need a protractor for this. Here is the plan for constructing a 90° angle at a point on a line.

Suppose you have a straight line and a point marked on it, and you want a line that goes straight up from that point to make a right angle. Follow these steps.

1. Place the point of your compasses on the marked point, open them to any convenient radius, and draw a semicircle (an arc) that crosses the line on both sides of the point. This cuts the line at two points, one to the left and one to the right. Call them A and B.

2. Put the compass point on A, open the compasses a little wider than before (more than half the distance between A and B), and draw a large arc above the line.

3. Without changing the compass opening, put the compass point on B and draw another large arc above the line. The two arcs cross each other at a point. Call it C.

4. Draw a straight line from the original marked point up to C. This new line is perpendicular to the original line, so the angle between them is exactly 90°.

Why does this work? The two arcs are drawn with the same compass opening, so AC and BC are equal lengths, and A and B are equally spaced either side of the point. The line through the middle therefore cuts the original line at a right angle. In the worked example below you will see the same method again, step by step.

WORKED_EXAMPLE

Worked example: Construct a 90° angle at the point P on a given line.

Given: A straight line with a point P marked on it.

Step 1. Place the compass point on P and open the compasses to a convenient radius. Draw a semicircle that cuts the line at A (on the left of P) and B (on the right of P). Points A, P and B all lie on the line, with P exactly between A and B.

Step 2. Widen the compasses a little (more than half the length of AB). Put the point on A and draw a large arc above the line.

Step 3. Keep the compass opening exactly the same. Move the point to B and draw another large arc above the line. This second arc crosses the first at point C.

Step 4. Use your ruler to draw a straight line from P through C, extending it a little beyond C. The angle between the original line and line PC is a right angle, 90°.

Check: Place the corner of a set square against the original line and slide it so its corner sits on P. The upright edge of the set square should lie exactly along the line you drew to C. That confirms the angle is 90°.

1. Draw a horizontal straight line and mark a point P somewhere along its middle. Using only a ruler and compasses, construct a line through P that is perpendicular to the horizontal line. Describe in your own words the four steps you used, and state the size of the angle you made.

answer: The line constructed through P stands straight up (perpendicular) from the horizontal line, and the angle between the two lines is exactly 90°.

2. You want to construct a 90° angle at point Q on a line. After drawing the semicircle from Q, the arc cuts the line at points A and B. What must be true about the compass opening you use when drawing the two crossing arcs from A and from B?

answer: The compass opening must be the same for the arc drawn from A and the arc drawn from B. (It must also be wider than half the distance from A to B so that the two arcs actually cross above the line.)

3. After following the construction steps you end up with point C, where the two arcs cross above the line. You then draw a straight line from P through C. Explain why this line makes a 90° angle with the original line, referring to the positions of A and B.

answer: A and B are equally spaced on either side of P (because P is the centre of the semicircle), and C is equally far from A and from B (because both arcs were drawn with the same opening). So the line through P and C runs exactly through the middle of A and B, straight up, which cuts the original line at a right angle of 90°.

Building a 60° Angle with Compass and Ruler

EXPLANATION

An angle of 60° turns up everywhere in construction, and it is beautifully easy to build because of a secret: an equilateral triangle has three equal sides and three equal angles, and each of those angles is exactly 60°. So if you can draw a triangle with all three sides the same length, you have built a 60° angle for free.

All you need is a pencil, a ruler, and a pair of compasses. The trick is that the compass can draw two arcs with exactly the same radius (the distance between its point and its pencil), and two equal radii give you two equal sides. Draw a third side between the two arc points and all three lengths match.

Here is the idea in words before we mark it out:

Draw a straight line and mark a point A on it. From A, the line going off to the right is one arm of your 60° angle.
Put the compass point on A, open it to any comfortable width, and swing an arc that crosses the line. Call where it lands on the line C. The distance A to C is your chosen radius.
Keep the compass exactly as wide as it is — do not change the opening — and put the point on C. Swing a second arc of the same radius so it crosses the first arc. Call the crossing point D.
Use your ruler to draw a straight line from A through D. The angle between the line AC and the line AD is 60°.

The reason this works is that AC, CD, and AD are all the same length (the first two were drawn with the same compass radius, and AD joins A to the meeting point of two equal-radius arcs). Three equal sides make an equilateral triangle, and every angle of an equilateral triangle is 60°.

WORKED_EXAMPLE

Worked example: Construct a 60° angle at point A on a given line.

Step 1. Draw a horizontal line and mark a point A on it. The line running to the right of A will be the base arm of your angle.

Step 2. Place the compass point on A. Open the compass to a comfortable width (say about 4 cm). Swing an arc that cuts across the line to the right of A. Label the point where the arc meets the line as C.

Step 3. This is the key step: without changing the compass opening at all, move the point of the compass onto C. Swing a second arc with the same radius so that it crosses the first arc above the line. Label the crossing point D.

Step 4. Draw a straight line from A through D using your ruler, extending it a little beyond D.

Step 5. The angle between the line AC and the line AD is the one you built. It measures 60°.

Why it is 60°: AC is the first compass radius, CD is the second compass radius, and because you kept the opening unchanged, AC = CD. The point D lies on both arcs, so AD is also the same radius, meaning AC = CD = AD. A triangle with three equal sides is equilateral, and each angle in an equilateral triangle is 60°. So the angle CAD is exactly 60°.

One handy tip: if you ever need a 60° angle but your base line only runs one way, just think of the base as any one side of an equilateral triangle — the construction works no matter which way the line points.

1. A learner draws a line and marks a point A on it. They place the compass point on A, open it, and swing an arc that cuts the line at C. Next they move the compass point to C and, keeping the opening unchanged, swing a second arc that crosses the first at D. They join A to D with a straight line. (a) Name the triangle that the three points A, C and D form. (b) Explain why the three sides AC, CD and AD must all be equal in length. (c) What is the size of the angle CAD, and why?

answer: (a) Triangle ACD is an equilateral triangle. (b) AC is the first compass radius and CD is drawn with the same opening, so AC = CD. The point D lies on both arcs, both drawn with the same radius, so AD is the same length too. Therefore AC = CD = AD. (c) Angle CAD is 60°, because an equilateral triangle has three equal angles and their total is 180°, so each one is 180° ÷ 3 = 60°.

2. You are going to construct a 60° angle at point A on a line. Put these steps in the correct order: 1. Keeping the compass opening the same, put the point on C and swing a second arc to cross the first, marking the crossing point D. 2. Draw a straight line from A through D. 3. Place the compass point on A and swing an arc that cuts the line, marking the point C. 4. Draw a straight line and mark a point A on it. Write the steps in order, and state the size of the angle you have built.

answer: The correct order is: 4, 3, 1, 2. First draw the line and mark A (4), then swing an arc from A to cut the line at C (3), then from C with the same opening swing a second arc to get D (1), and finally join A to D (2). The angle CAD built is 60°.

3. In the construction of a 60° angle, the three marked points A, C and D form an equilateral triangle, so each of its angles is 60°. (a) What is the total size of the three angles of any triangle? (b) Using your answer to (a), work out the size of one angle of an equilateral triangle. (c) A student instead draws an arc from A of radius 3 cm, then from C swings an arc of radius 5 cm. Can this still produce a 60° angle? Explain your answer.

answer: (a) The three angles of any triangle add up to 180°. (b) 180° ÷ 3 = 60°, so each angle is 60°. (c) No. If the two radii are different, then AC ≠ CD, so the triangle is not equilateral and the angle at A is not 60°. The 60° construction only works when both arcs use exactly the same radius.

C1 `The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Construction

`subtopic`

Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

`term`

Third Term

`week`

6

`lesson_objectives`

• construct parallel lines
• construct perpendicular lines
• bisect a given line segment
• construct angle 90°
• construct angle 60°

`entry_behaviour`

• Students have been taught how to draw lines.

`instructional_materials`

• Mathematical set.

Objective 1 — construct parallel lines

`objective_text`

construct parallel lines

`current_step`

1

`instructional_events`

• present_content
• elicit_performance

`student_tasks`

• construct parallel lines, following the teacher's board construction and replicating it in their notes

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — construct perpendicular lines

`objective_text`

construct perpendicular lines

`current_step`

2

`instructional_events`

• present_content
• elicit_performance

`student_tasks`

• follow the teacher's board construction and replicate it in their notes

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 3 — bisect a given line segment

`objective_text`

bisect a given line segment

`current_step`

3

`instructional_events`

• present_content
• elicit_performance

`student_tasks`

• construct line segment, following the teacher's board construction and replicating it in their notes

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 4 — construct angle 90°

`objective_text`

construct angle 90°

`current_step`

4

`instructional_events`

• present_content
• elicit_performance

`student_tasks`

• construct angle 90 degrees, following the teacher's board construction and replicating it in their notes

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 5 — construct angle 60°

`objective_text`

construct angle 60°

`current_step`

5

`instructional_events`

- present_content
- elicit_performance

`student_tasks`

- construct angle 60 degrees, following the teacher's board construction and replicating it in their notes

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Drawing Parallel Lines

EXPLANATION

Two lines are parallel when they run side by side and never meet, no matter how far they are extended. In this section you will draw your own pair of parallel lines using just a set square and a ruler.

Here is the method, step by step.

Step 1 — Draw the first line. Use your ruler to draw a straight line. Label its ends A and B. This is the line your new line must run parallel to.

Step 2 — Seat the set square on the line. Place your set square so that one of its straight edges lies exactly along the line AB. Hold it firmly so it does not slip.

Step 3 — Bring the ruler to the set square. Press the ruler firmly against another edge of the set square, so that the set square and the ruler touch along that whole edge. Hold the ruler absolutely still.

Step 4 — Slide the set square. Keeping the ruler fixed in place, slide the set square along the ruler to wherever you want the second line to be. Because the set square is guided by the ruler, the edge that lay on AB always keeps the same direction as it slides.

Step 5 — Draw the second line. Draw a straight line along the edge of the set square that was first on AB. Label its ends C and D. The line CD is parallel to AB.

The key to the whole construction is the ruler: it never moves, so it keeps the set square turning at the same angle all the way along. That is what makes the two lines point the same way.

When you check your work, the two lines should be the same distance apart everywhere along their length — the gap at the left end should match the gap at the right end.

WORKED_EXAMPLE

Worked example. Construct a line parallel to the given line PQ.

1. With your ruler, draw the line PQ.
2. Lay one edge of the set square exactly on PQ and hold it in place.
3. Press the ruler against another edge of the set square, then hold the ruler still.
4. Slide the set square along the ruler to a new position, about 4 cm above PQ.
5. Draw a line along the edge that was on PQ, and label it RS.

The line RS is parallel to PQ: both are straight, they point the same way, and they never meet.

1. Draw a line and label it XY. Using your set square and ruler, construct a line parallel to XY, and label it UV. How can you check that your two lines really are parallel?

answer: Use a ruler to measure the gap between XY and UV in two places — near the left end and near the right end. If the two gaps are the same, the lines are parallel.

2. Construct two parallel lines by sliding your set square along the ruler to a second position. If you slide the set square two centimetres further along the ruler before drawing the second line, are the two lines still parallel? Explain why.

answer: The two lines are parallel because the ruler held the set square at the same angle the whole way, so CD keeps the same direction as AB. They point the same way and never meet, so they are a pair of parallel lines.

Constructing Perpendicular Lines

EXPLANATION

Two lines are perpendicular when they cross each other at a right angle — that is, at an angle of exactly 90°. You can see this in the corners of a page or a ruler. In construction, we can draw a line perpendicular to another one using just a ruler and a set square.

Your set square is a triangular tool with one corner that is exactly a right angle. That right-angle corner is your tool for drawing perpendicular lines. The key idea is this: if you slide the set square along a ruler, every line you draw along the set square's edge makes the same angle with the ruler — a right angle.

WORKED_EXAMPLE

Let's draw a line perpendicular to the line AB.

Step 1 — Place your ruler firmly along AB. Do not let it move.

Step 2 — Set the right-angle corner of your set square against the ruler, with the set square resting on the ruler's edge. One arm of the set square lies along the ruler.

Step 3 — Hold the ruler still and slide the set square along it until the other arm of the set square passes through the point where you want the perpendicular line. If you are drawing the perpendicular through a given point on AB, slide the set square until that point lies along the set square's edge.

Step 4 — Keeping the set square pressed against the ruler, draw a line along the set square's edge. Because the set square's corner is a right angle, this new line makes 90° with AB.

Step 5 — Label the new line CD. AB and CD are perpendicular, which you write as AB ⊥ CD. Check your work: the two lines meet at a right angle, like the corner of a page.

1. Draw a line PQ on your page. Use your set square and ruler to construct a line that is perpendicular to PQ, passing through a point M on PQ. Which angle do the two lines make where they cross, and how do you write that the two lines are perpendicular?

answer: The two lines make a right angle of 90°. You write PQ ⊥ the new line (for example, PQ ⊥ MN).

2. You construct a line perpendicular to AB. When you measure the angle where the new line meets AB, which of these should you find: 30°, 60°, 90°, or 180°? Explain why your set square is the tool that guarantees this.

answer: You should find 90°. Your set square has a corner that is exactly a right angle, so a line drawn along its edge while the set square is held against the ruler makes a 90° angle with the line along the ruler.

Cutting a Line into Two Equal Parts

EXPLANATION

To bisect a line segment means to cut it into two equal halves. The point where the cut happens is the midpoint of the segment. You already know how to draw a straight line; now you will use your mathematical set — a ruler and a pair of compasses — to find the exact middle of a line segment and mark it.

Here is how to bisect a line segment AB, drawn with your ruler.

1. Set your compasses to a radius a little more than half the length of AB. Keep this opening the same for every step below — do not change it.
2. Place the point of the compasses on A and draw an arc above the line, and another arc below the line. (Each arc is a curve that swings over the line.)
3. Move the compasses to B — without changing the opening — and draw two more arcs, one above and one below the line. Make sure these new arcs cross the first two.
4. Now look above the line: where the two arcs cross, that is one point, call it P. Below the line, where the two arcs cross, that is another point, call it Q.
5. Draw a straight line through P and Q with your ruler. This line crosses AB. The point where it crosses AB is the midpoint M, and it splits AB into two equal parts: AM = MB.

Your compasses did the real work for you. Because every arc was drawn with the same opening, every point on an arc is the same distance from its centre. So P is exactly as far from A as it is from B (both distances are the compass opening), and Q is exactly as far from A as it is from B too. A point that is equally distant from A and B lies exactly above the middle of AB, so the line through P and Q passes straight through the midpoint. This is why your construction lands exactly on the middle every time, not just close to it.

WORKED_EXAMPLE

Let us bisect a line segment AB by hand, step by step.

Draw AB with your ruler, say 8 cm long.

Open your compasses to a little more than half of 8 cm — about 5 cm. Keep that opening fixed.

From A, draw an arc above the line and an arc below the line. Then, keeping the compasses at the same 5 cm opening, from B draw another arc above and another arc below, so the four arcs cross. Mark the crossing point above the line as P and the crossing point below the line as Q.

Lay your ruler along P and Q and draw a straight line through them. Where this line meets AB is the midpoint M. Measure AM and MB with your ruler: each is 4 cm, exactly half of 8 cm. The segment AB has been bisected.

The line through P and Q is special: it not only splits AB into two equal halves, it also meets AB at a right angle. When you learn how to construct perpendicular lines, you will see that you have just drawn one without meaning to.

1. Draw a line segment CD of length 6 cm. Using your compasses and ruler only, bisect it. Label the midpoint M. What are the lengths of CM and MD?

answer: CM = 3 cm and MD = 3 cm.

2. When bisecting a line segment, why must you keep the same compass opening for all four arcs?

answer: So that every point on an arc is the same distance from its centre. The crossing points P and Q are then equally distant from both ends of the segment, which places them directly above the midpoint, so the line through them lands exactly on the middle.

Constructing an Angle of 90°

EXPLANATION

You already know how to draw straight lines. Now you will learn to construct a right angle — an angle of exactly 90° — using only your compass and ruler. You will not measure it with a protractor; you will build it with arcs.

To construct a 90° angle at a point on a line, you first make a semicircular arc, then cut that arc in half. Where the two cuts cross gives you the point that joins back to your starting point to form the right angle. Let's see this in steps.

WORKED_EXAMPLE

Construct an angle of 90° at the point P on a straight line.

Step 1 — Draw a straight line and mark a point P on it. Put the point of your compass on P.

Step 2 — Open your compass to any convenient width and draw a semicircular arc that crosses the line on both sides of P. Call the two points where the arc meets the line A and B.

Step 3 — Put the point of your compass on A. Keeping the same opening, draw an arc above the line.

Step 4 — Without changing the opening, put the point of your compass on B and draw another arc above the line. The two arcs cross at a point. Call it Q.

Step 5 — Draw a straight line from P through Q. The angle between the line PQ and your original line is 90°.

Why this works: A and B are equal distances from P, so P is the middle of the semicircular arc. Because you kept the same compass opening, the arc from A and the arc from B are equal, and they cross exactly midway over the arc. The point Q therefore sits straight above P, so the line P–Q stands perpendicular to your original line.

1. Construct an angle of 90° at the point M on a straight line. Draw your own straight line, mark a point M on it, and use your compass to build the right angle. Describe briefly the arcs you drew and the point where they crossed.

answer: Your finished drawing should show a line through M standing straight up from the original line, making a right angle of 90° between them. You drew a semicircular arc across the line from M, then two equal arcs above the line — one from each end of the semicircle — crossing at a point above M. The line from M through that crossing point is perpendicular to the original line.

2. A point R lies on a line, and you have already drawn a semicircular arc across the line from R, meeting it at C and D. Explain in your own words the next two steps to finish constructing a 90° angle at R, and state what compass opening you must use.

answer: Put the compass point on C and draw an arc above the line, then, without changing the opening, put the point on D and draw another arc above the line. The two arcs cross at a point above R; draw the line from R through that crossing. You must keep the same compass opening for both arcs — the same opening you used for the semicircle.

Constructing a 60° Angle

EXPLANATION

A 60° angle is one of the most useful angles to construct, because it forms the basis for drawing equilateral triangles and other shapes. Like the other constructions you have practised, you make it with just a ruler and a compass — no protractor needed.

The idea behind it is simple. An equilateral triangle has three equal sides and three equal angles, and each of those angles is exactly 60°. So if you build part of an equilateral triangle, you have constructed a 60° angle. The compass does the work: arcs drawn with the same radius mark out points that are equal distances apart, which is exactly what builds that equal-sided triangle.

Here is the construction in four steps.

Step 1 — Start with a ray. Draw a straight line and mark a point at the left end of it. Call this point A. This is the point where your 60° angle will sit, so it is called the *vertex*. The line from A to the right is one arm of your angle.

Step 2 — Strike an arc from the vertex. Put the sharp point of your compass on A and open it to any comfortable width. Draw an arc that crosses the line. Mark where the arc meets the line and call this point B.

Step 3 — Strike a matching arc from B. Here is the key move: do **not** change the width of your compass. Keep it exactly as it was in Step 2. Put the sharp point on B and draw another arc, cutting the first arc. Mark the point where the two arcs cross and call it C.

Step 4 — Join up the angle. Use your ruler to draw a straight line from A through C. The angle between the line AC and your original line AB is exactly 60°.

Because you never changed the compass width, the distance from A to B, from B to C, and from A to C are all equal — the three sides of an equilateral triangle. And since an equilateral triangle has equal angles, each of those angles is 60°. That is why your construction works.

WORKED_EXAMPLE

Let us watch the whole construction again, all at once, so you can see how the steps fit together.

1. Draw a line and mark a point A near the left end. This is the vertex of your 60° angle.
2. Put the compass point on A, open it to a convenient width, and draw an arc cutting the line. Call the cut point B.
3. Keep the compass width unchanged. Move the point to B and draw a second arc so that it crosses the first arc. Call the crossing point C.
4. Draw a straight line from A through C.

Check that you have three equal distances: AB = BC = AC. Because those three lengths are equal, triangle ABC is equilateral, so every angle in it — including the angle between AB and AC at A — is 60°.

The single most important thing to remember is to keep the compass opening the same from Step 2 through Step 3. If you change it, the sides are no longer equal and your angle will not be exactly 60°.

1. Follow the construction to build a 60° angle on your paper, then describe in your own words the two moves you make with the compass.

answer: You draw a line with a vertex A, strike one arc from A cutting the line at B, then — keeping the compass width unchanged — strike a second arc from B cutting the first at C, and join A to C. The angle between AB and AC is 60°.

2. When you construct a 60° angle, why must you keep the compass width the same for the second arc as for the first? What would happen if you changed it?

answer: Keeping the same width makes the three distances AB, BC and AC all equal, so the figure is an equilateral triangle and each angle is 60°. If you changed the width, the sides would no longer be equal, so the angle would not be exactly 60°.

3. You have drawn a ray from point A and struck an arc from A that cuts the ray at B. List the next three things you do to finish constructing a 60° angle.

answer: 1. Keep the compass at the same width and put its point on B. 2. Draw an arc from B that cuts the first arc, and mark the crossing point C. 3. Draw a straight line from A through C; the angle between that line and the ray is 60°.

C2 `C1 plus the lesson note as source evidence.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Construction

`subtopic`

Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

`term`

Third Term

`week`

6

`lesson_objectives`

• construct parallel lines
• construct perpendicular lines
• bisect a given line segment
• construct angle 90°
• construct angle 60°

`entry_behaviour`

• Students have been taught how to draw lines.

`instructional_materials`

• Mathematical set.

Objective 1 — construct parallel lines

objective_text

construct parallel lines

current_step

1

instructional_events

• present_content
• elicit_performance

student_tasks

• construct parallel lines, following the teacher's board construction and replicating it in their notes

source_figures


```
[
 {
  "caption": "Line $AB$, with a point $P$ marked above it. A set square is drawn below $AB$ with
one edge along the line: the right angle is at the left end of that edge, one leg runs down from
it, and the hypotenuse closes back to a point on $AB$ to the right.",
  "figure_ref": "fig-01"
 },
 {
  "caption": "The same set square on $AB$, with the ruler now drawn as a long strip lying along
the set square's downward edge. The set square's vertical edge and the ruler's right edge are
**layered** — only one line is visible on the page. The right angle marked at the corner is the
set square's own, between the edge lying along $AB$ and the edge rising from it.",
  "figure_ref": "fig-02"
 },
 {
  "caption": "The same arrangement drawn a second time alongside the first — the set square's
edge again **layered** on the ruler's right edge, with no gap between them — and with the ruler
leaning — running from upper left to lower right across $AB$ instead of standing vertical. Only
$A$ is labelled. The slanted ruler is the point of this panel — it shows that the ruler's
inclination does not affect the construction.",
  "figure_ref": "fig-03"
 },
 {
  "caption": "The vertical ruler held fixed, with the set square drawn twice — once with its edge
on $AB$ and once higher up, its edge on a second line parallel to $AB$ — and an arrow between the
two positions pointing up along the ruler.",
  "figure_ref": "fig-04"
 },
 {
  "caption": "The same slide drawn a second time alongside it, with the leaning ruler. Both set
square positions are **layered** on the ruler's right edge, as in fig-03. Here the upper line
carries the point $P$, and the set square's upper position has its edge running through $P$.",
  "figure_ref": "fig-05"
 }
]
```


lesson_content

```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Examples:** construct a line through Point P so that it is parallel to AB.\n\n(a)
Place a set square so that one edge is accuratly along AB\n\n**fig-01.** Line $AB$, with a point
$P$ marked above it. A set square is drawn below $AB$ with\none edge along the line: the right
angle is at the left end of that edge, one leg runs down from\nit, and the hypotenuse closes back
to a point on $AB$ to the right.\n\n[drawing attached to this section as fig-01, marked A, B, P]
\n\n(b) Place a ruler along one of the other edge of the set square. Use the left hand edge if
you\nare right handed.\n\n**fig-02.** The same set square on $AB$, with the ruler now drawn as a
long strip lying along the\nset square's downward edge. The set square's vertical edge and the
ruler's right edge are\n**layered** — only one line is visible on the page. The right angle
marked at the corner is the\nset square's own, between the edge lying along $AB$ and the edge
rising from it.\n\n[drawing attached to this section as fig-02, marked A, B]\n\n**fig-03.** The
same arrangement drawn a second time alongside the first — the set square's edge\nagain
**layered** on the ruler's right edge, with no gap between them — and with the ruler leaning\n—
running from upper left to lower right across $AB$ instead of standing vertical. Only $A$
is\nlabelled. The slanted ruler is the point of this panel — it shows that the ruler's
inclination\ndoes not affect the construction.\n\n[drawing attached to this section as fig-03,
marked A]\n\n(c) Hold the ruler firmly, slide the set square along the ruler towards P. Stop when
the edge that\nwas on AB reaches P. Draw a line along this edge of the set square through P.
\n\n**fig-04.** The vertical ruler held fixed, with the set square drawn twice — once with its
edge on\n$AB$ and once higher up, its edge on a second line parallel to $AB$ — and an arrow
between the two\npositions pointing up along the ruler.\n\n[drawing attached to this section as
fig-04, marked A, B]\n\n**fig-05.** The same slide drawn a second time alongside it, with the
leaning ruler. Both set\nsquare positions are **layered** on the ruler's right edge, as in
fig-03. Here the upper\nline carries the point $P$, and the set square's upper position has its
edge running through $P$.\n\n[drawing attached to this section as fig-05, marked A, P]"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Objective 2 — construct perpendicular lines

objective_text

construct perpendicular lines

current_step

2

instructional_events

• present_content
• elicit_performance

student_tasks

• follow the teacher's board construction and replicate it in their notes

source_figures


```
[
 {
  "caption": "Line $AB$ with a second line $XY$ crossing it, $X$ above and $Y$ below, and a
right-angle mark in the upper right quadrant at the crossing.",
  "figure_ref": "fig-06"
 },
 {
  "caption": "A plain line with a stroke at each end, labelled $A$ on the left and $B$ on the
right. The point $Q$ named in the instruction is not marked on the drawing.",
  "figure_ref": "fig-07"
 },
 {
  "caption": "Line $AB$ with the set square standing on it as a narrow right-angled triangle: a
short base along $AB$, a near-vertical edge rising from the right end of that base, and the
hypotenuse running up from the left.",
  "figure_ref": "fig-08"
 },
 {
  "caption": "Line $AB$ with $XY$ now drawn across it, $X$ above and $Y$ below, and a right-angle
mark in the upper right quadrant. The foot of $XY$ is not labelled $Q$.",
  "figure_ref": "fig-09"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "Two lines are said to be perpendicular to each other when they meet at a right
angle i.e\n$90^\\circ$\n\n**fig-06.** Line $AB$ with a second line $XY$ crossing it, $X$ above
and $Y$ below, and a\nright-angle mark in the upper right quadrant at the crossing.\n\n[drawing
attached to this section as fig-06, marked A, B, X, Y]\n\n1. Draw a line AB mark a point Q on the
line.\n\n**fig-07.** A plain line with a stroke at each end, labelled $A$ on the left and $B$ on
the right.\nThe point $Q$ named in the instruction is not marked on the drawing.\n\n[drawing
attached to this section as fig-07, marked A, B]\n\n2. Place the edge of the set square on line
AB. Let the edge of the ruler meet the hypotenuse of\n   the set square as shown.\n\n**fig-08.**
Line $AB$ with the set square standing on it as a narrow right-angled triangle: a\nshort base
```

```
along $AB$, a near-vertical edge rising from the right end of that base, and the\nhypotenuse
running up from the left.\n\n[drawing attached to this section as fig-08, marked A, B]\n\n3.
Pressing firmly on the ruler, slide the set square down along the hypotenuse side until the\n
third edges touches point Q. Draw a line XY\n\n**fig-09.** Line $AB$ with $XY$ now drawn across
it, $X$ above and $Y$ below, and a right-angle\nmark in the upper right quadrant. The foot of
$XY$ is not labelled $Q$.\n\n[drawing attached to this section as fig-09, marked A, B, X, Y]"
 }
]
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Objective 3 — bisect a given line segment

objective_text

bisect a given line segment

current_step

3

instructional_events

• present_content
• elicit_performance

student_tasks

• construct line segment, following the teacher's board construction and replicating it in their notes

source_figures


```
[
 {
  "caption": "The line segment $AB$, a stroke at each end, $A$ on the left and $B$ on the
right.",
  "figure_ref": "fig-10"
 },
 {
  "caption": "The segment $AB$ with arc pairs crossing above and below it, the upper crossing
labelled $P$ and the lower one $Q$. Each crossing is drawn as a small X where an arc from $A$
meets an arc from $B$.",
  "figure_ref": "fig-11"
 },
 {
  "caption": "The same arcs with $P$ joined to $Q$, the joining line crossing $AB$ at its middle
and carrying a right-angle mark in the upper right quadrant.",
  "figure_ref": "fig-12"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Procedures:**\n\n1. Open a pair of compasses so that the radius is about $\
\frac{3}{4}$ of the length of AB.\n2. To bisect the line segment AB means to divide it into two
equal halves\n\n**fig-10.** The line segment $AB$, a stroke at each end, $A$ on the left and $B$
on the right.\n\n[drawing attached to this section as fig-10, marked A, B]\n\n3. Place the sharp
point of the compass on A, Draw two arcs, one above and the other below the\n   midle of AB.
\n\n**fig-11.** The segment $AB$ with arc pairs crossing above and below it, the upper
crossing\nlabelled $P$ and the lower one $Q$. Each crossing is drawn as a small X where an arc
from $A$ meets\nan arc from $B$.\n\n[drawing attached to this section as fig-11, marked A, B, P,
Q]\n\n4. With the same radius, and place the sharp point of the compasses on B Draw two arc so
that they\n   cut the first arc at P and Q\n\n**fig-12.** The same arcs with $P$ joined to $Q$,
the joining line crossing $AB$ at its middle and\ncarrying a right-angle mark in the upper right
quadrant.\n\n[drawing attached to this section as fig-12, marked A, B, P, Q]"
```

```
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Objective 4 — construct angle 90°

`objective_text`

construct angle 90°

`current_step`

4

`instructional_events`

• present_content
• elicit_performance

`student_tasks`

• construct angle 90 degrees, following the teacher's board construction and replicating it in their notes

`source_figures`


```
[
 {
  "caption": "The line $AB$ with a stroke at each end, the two arc cuts $X$ and $Y$ marked either side of the middle, and the perpendicular $OZ$ erected from $O$ between them, with a right-angle mark at its foot and the two arcs crossing at $Z$ near its top.",
  "figure_ref": "fig-13"
 }
]
```


`lesson_content`


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Procedure:**\n\n1. Draw a line segment $AB$ and mark a point $O$ on it where the right angle is to stand.\n2. With centre $O$ and any convenient radius, draw an arc to cut $AB$ at $X$ and $Y$.\n3. With centre $X$ and a radius greater than $XO$, draw an arc above $AB$.\n4. With centre $Y$ and the same radius as in (3), draw an arc to cut the first arc at $Z$.\n5. Join $OZ$. Then $\\angle ZOA = \\angle ZOB = 90^\\circ$.\n\n**fig-13.** The line $AB$ with a stroke at each end, the two arc cuts $X$ and $Y$ marked either\nside of the middle, and the perpendicular $OZ$ erected from $O$ between them, with a right-angle\nmark at its foot and the two arcs crossing at $Z$ near its top.\n\n[drawing attached to this section as fig-13, marked A, B, X, Y, O, Z]"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Objective 5 — construct angle 60°

objective_text

construct angle 60°

current_step

5

instructional_events

• present_content
• elicit_performance

student_tasks

• construct angle 60 degrees, following the teacher's board construction and replicating it in their notes

source_figures


```
[
 {
  "caption": "The line $BC$ with $B$ at the left, a stroke marking $X$ part way along, a long arc swung from $B$ through $X$ and up to the left, a shorter arc swung from $X$ cutting it, and a straight line from $B$ to that crossing. The crossing itself is not labelled.",
  "figure_ref": "fig-14"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Procedure:**\n\n1. Draw a line segment BC with centre B at any convinent radius. Draw an arc to cut BC at X\n2. with centre X and the same radius as in (i) draw an arc to cut the first arc at A\n3. Join AB.\n\n**fig-14.** The line $BC$ with $B$ at the left, a stroke marking $X$ part way along, a long arc\nswung from $B$ through $X$ and up to the left, a shorter arc swung from $X$ cutting it, and a\nstraight line from $B$ to that crossing. The crossing itself is not labelled.\n\n[drawing attached to this section as fig-14, marked B, C, X]"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Constructing a line parallel to a given line

EXPLANATION

Two lines are **parallel** when they are always the same distance apart, so they never meet, no matter how far they are extended. In this section you will learn to draw a line through a given point that runs parallel to a line you already have. You will need your **set square** and your **ruler**.

Here is the task. You are given a line AB and a point P above it, and you want a straight line through P that is parallel to AB (fig-01). The set square is the key tool, because its edges make a right angle, and the sliding trick below keeps that angle fixed while you move.

Step 1 — Lay the set square on the line. Place the set square so that one of its edges lies exactly along AB (fig-01). It does not matter which end of the set square is which, or how big the set square is — the only thing that matters is that one edge sits squarely on the line.

Step 2 — Rest the ruler against another edge. Place your ruler along one of the set square's *other* edges (fig-02). If you are right-handed, use the edge on your left hand side, because it is easier to hold steady. In fig-03 the ruler rests against the set square's sloping edge instead; either edge works, and the ruler's own slope does not affect the construction at all.

Step 3 — Hold the ruler still and slide the set square. Keep the ruler pressed firmly in place with one hand, so it does not move. With the other hand, slide the set square along the ruler towards P, stopping as soon as the edge that was lying on AB reaches P (fig-04 and fig-05). Sliding the set square along a held ruler moves it without turning it, so its edges stay at the same angle throughout.

Step 4 — Draw the parallel line. Keeping the set square pressed against the ruler, draw a straight line along the edge that passes through P. Because that edge has kept exactly the same direction it had when it lay on AB, the new line through P is parallel to AB.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above right] at (8,0) {$B$};
  \draw (0.35,2.2) -- (8,2.2);
  \fill (6.0,2.2) circle (1.6pt);
  \node[above] at (6.0,2.2) {$P$};
  % The ruler is held still; it spans both positions of the set square, drawn a hairline
  % clear of their sloping edges.
  \draw[rounded corners=2pt] (-0.026,-2.119) -- (7.616,2.247) -- (7.824,1.882) -- (0.182,-2.484) -- cycle;
  \draw (0.6,0) -- (0.6,-1.6) -- (3.4,0);
  \draw (0.6,-0.26) -- (0.86,-0.26) -- (0.86,0);
  % The same set square, translated along the ruler until its edge reaches P.
  \draw (4.45,2.2) -- (4.45,0.6) -- (7.25,2.2);
  \draw (4.45,1.94) -- (4.71,1.94) -- (4.71,2.2);
  \draw[->] (2.6,0.55) -- (4.0,1.35);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above] at (8,0) {$B$};
  \draw (0.71,2.40) -- (8,2.40);
  \draw[rounded corners=3pt] (0.28,3.65) rectangle (0.71,-3.97);
  % ONE set square drawn in two positions, translated straight up the ruler's right edge
  % (x=0.71). Both are congruent — leg 1.21, width 2.68 — so the two hypotenuses are
  % exactly parallel, which is the visible signature of a slide.
  \draw (0.71,0) -- (0.71,-1.21) -- (3.39,0);
  \draw (0.71,-0.26) -- (0.97,-0.26) -- (0.97,0);
```

```
  \draw (0.71,2.40) -- (0.71,1.19) -- (3.39,2.40);
  \draw (0.71,2.14) -- (0.97,2.14) -- (0.97,2.40);
  \draw[->] (2.44,0.1) -- (2.44,1.37);
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above right] at (8,0) {$B$};
  % The ruler now lies along the sloping edge, extending past it at both ends. Drawn a
  % hairline clear of that edge so the set square stays visible as a triangle.
  \draw[rounded corners=2pt] (-0.171,-3.033) -- (6.760,0.545) -- (6.967,0.145) -- (0.036,-3.433) -- cycle;
  \draw (1.72,0) -- (1.72,-1.9) -- (5.4,0);
  \draw (1.72,-0.26) -- (1.98,-0.26) -- (1.98,0);
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above] at (8,0) {$B$};
  \draw[rounded corners=3pt] (1.29,3.50) rectangle (1.72,-4.12);
  % The set square's vertical edge lies ALONG the ruler's right edge (x=1.72) — in the
  % source the two are layered, so only one line is visible. Drawn collinear, not offset.
  \draw (1.72,0) -- (1.72,-1.9) -- (5.4,0);
  \draw (1.72,-0.26) -- (1.98,-0.26) -- (1.98,0);
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \node[left] at (0,0) {$A$};
  \node[above] at (7,0) {$B$};
  \fill (3.4,2.8) circle (1.6pt);
  \node[above] at (3.4,2.8) {$P$};
  \draw (2.7,0) -- (2.7,-1.1) -- (5.2,0);
  \draw (2.7,-0.26) -- (2.96,-0.26) -- (2.96,0);
\end{tikzpicture}
```

```
WORKED_EXAMPLE
**Example.** Construct a line through point $P$ so that it is parallel to the given line $AB$.

Start with your set square flat on the page and slide one edge until it lies exactly along $AB$ (fig-01). Now press your ruler against one of the set square's other edges, for example its sloping edge, as in fig-03. Hold the ruler firmly so it cannot slip. Slide the set square up along the ruler, without turning it, until the edge that was on $AB$ passes exactly through $P$ (fig-05). Finally, draw a straight line along that edge through $P$. The line you have drawn is parallel to $AB$, because the edge was moved along the ruler without being turned, so it kept the same direction all the way.
```

1. Using your set square and ruler, construct a line through the point P that is parallel to the line AB. Describe the steps you take.
answer: Lay one edge of the set square exactly along AB; press the ruler against another edge of the set square and hold it firmly; slide the set square along the ruler without turning it until the edge that was on AB passes through P; draw a straight line along that edge through P. This line is parallel to AB.

2. Why must you hold the ruler firmly while you slide the set square? What would happen if the ruler moved?
answer: The ruler must not move so that the set square slides along it without turning. If the ruler slipped, the set square would change direction as it moved, and the line you draw through P would not be parallel to AB.

Constructing Perpendicular Lines

EXPLANATION

Two lines are said to be perpendicular to each other when they meet at a right angle — that is, an angle of 90°.

Look at fig-06. Line AB runs from left to right, and a second line XY crosses it, with X above and Y below. At the crossing there is a small square mark in the upper right quadrant, showing that the two lines meet at a right angle. Because they meet at 90°, line XY is perpendicular to line AB.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (-0.5,0) -- (7.5,0);
  \draw (0,0.25) -- (0,-0.25);
  \draw (5,0.25) -- (5,-0.25);
  \node[below] at (0,-0.3) {$A$};
  \node[below] at (5,-0.3) {$B$};
  \draw (2.5,2.6) -- (2.5,-2.6);
  \node[above] at (2.5,2.6) {$X$};
  \node[below] at (2.5,-2.6) {$Y$};
  \draw (2.82,0) -- (2.82,0.32) -- (2.5,0.32);
\end{tikzpicture}
```

WORKED_EXAMPLE

Let's construct a line perpendicular to a given line AB through a point Q on it.

Step 1. Draw a line AB and mark a point Q on it, as in fig-07.

Step 2. Place the edge of the set square on line AB. Let the edge of the ruler meet the hypotenuse of the set square, as in fig-08. The upright edge of the set square stands square on line AB, with a right angle marked where it meets the base.

Step 3. Pressing firmly on the ruler, slide the set square down along the hypotenuse side until the third edge touches point Q. Then draw a line XY.

In fig-09 you can see the result: line XY crosses AB at Q, with X above and Y below, and a right-angle mark in the upper right quadrant at the crossing. Since XY meets AB at 90°, the line XY you have drawn is perpendicular to AB.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below] at (0.15,-0.1) {$A$};
  \node[above] at (4.52,0.1) {$B$};
  \draw (2.1,1.85) -- (2.15,-0.8);
  \node[above] at (2.1,1.85) {$X$};
  \node[below] at (2.15,-0.8) {$Y$};
  \draw (2.44,0) -- (2.44,0.32) -- (2.12,0.32);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below] at (0.41,-0.05) {$A$};
  \node[above] at (3.93,0.05) {$B$};
  % The upright edge stands square on the base: step 2 and 3 name a hypotenuse and a
  % "third edge", which only a marked right angle lets the reader tell apart.
  \draw (1.54,0) -- (2.25,1.0) -- (2.25,0) -- cycle;
  \draw (1.97,0) -- (1.97,0.28) -- (2.25,0.28);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
```

```
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below] at (0,-0.28) {$A$};
  \node[below] at (7,-0.28) {$B$};
\end{tikzpicture}
```

1. Look at a point Q marked on a line AB. What is the angle between line AB and a line drawn perpendicular to it at Q?

answer: 90° — a right angle.

2. Describe the steps you would follow to construct a line perpendicular to a given line AB through a point Q on it, using a set square and ruler.

answer: Place the edge of the set square on line AB with the edge of the ruler meeting the hypotenuse. Pressing firmly on the ruler, slide the set square along the hypotenuse until the third edge touches point Q, then draw a line through Q.

Bisecting a Line Segment

EXPLANATION

To bisect a line segment means to divide it into two equal halves. So if you bisect the segment AB, you split it at a point M that lies exactly halfway along it, so that $AM = MB$.

You only need a ruler, a pencil and a pair of compasses. Here is the whole construction.

1. Draw the line segment AB with a ruler, marking a short stroke across each end (fig-10).
2. Open the compasses so that the radius is about $\frac{3}{4}$ of the length of AB.
3. Place the sharp point of the compasses on A. With this opening, draw two arcs — one above the middle of AB and one below it.
4. Without changing the compass opening, move the sharp point to B. Draw two arcs again, one above and one below AB, so that each new arc cuts the matching arc from A. The arcs that cross above AB meet at a point you call P; the arcs that cross below AB meet at a point you call Q (fig-11).
5. Join P to Q with your ruler.
6. The line PQ crosses AB at M. This M is the midpoint of AB, so $AM = MB$. The line PQ also meets AB at 90° (fig-12).

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (6,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (6,0.22) -- (6,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (6,-0.1) {$B$};
  \draw (0,0) ++(37:4.5) arc (37:59:4.5);
  \draw (6,0) ++(121:4.5) arc (121:143:4.5);
  \draw (0,0) ++(-59:4.5) arc (-59:-37:4.5);
  \draw (6,0) ++(-143:4.5) arc (-143:-121:4.5);
  \draw (3,3.35) -- (3,-3.35);
  \node[above] at (3,3.35) {$P$};
  \node[below] at (3,-3.35) {$Q$};
  \node[below right] at (3,-0.1) {$M$};
  \draw (3.32,0) -- (3.32,0.32) -- (3,0.32);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (6,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (6,0.22) -- (6,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (6,-0.1) {$B$};
  \draw (0,0) ++(37:4.5) arc (37:59:4.5);
  \draw (6,0) ++(121:4.5) arc (121:143:4.5);
  \draw (0,0) ++(-59:4.5) arc (-59:-37:4.5);
  \draw (6,0) ++(-143:4.5) arc (-143:-121:4.5);
  \node[above] at (3,3.5) {$P$};
  \node[below] at (3,-3.5) {$Q$};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (7,-0.1) {$B$};
\end{tikzpicture}
```

WORKED_EXAMPLE

Worked example — bisecting a segment

Take a segment AB that is 8 cm long and bisect it.

Step 1. Draw AB, 8 cm long, with a stroke at each end.

Step 2. Open the compasses to a radius of about $\frac{3}{4}$ of 8 cm, which is about 6 cm. This opening must stay the same for the whole construction.

Step 3. Put the point on A and draw one arc above the middle of AB and one below it.

Step 4. Keep the same opening, put the point on B, and draw two arcs that cut the arcs from A. Label the crossing above AB as P and the crossing below as Q.

Step 5. Join P to Q with the ruler.

Step 6. Where PQ meets AB, label that point M. Reading the drawing, M sits at 4 cm along the segment, so $AM = 4\text{ cm}$ and $MB = 4\text{ cm}$ — the segment is cut into two equal halves. PQ also meets AB at a right angle.

1. Draw a line segment AB that is 6 cm long and bisect it with your compasses and ruler. Use P for the crossing of the arcs above AB and Q for the crossing below.

answer: The segment is cut into two equal halves: M is the midpoint of AB, so $AM = MB = 3$ cm. The line PQ also meets AB at 90°.

Constructing an angle of 90°

EXPLANATION

An angle of 90° — a right angle — is the angle between a line and a perpendicular raised to it. You can build it exactly with your compasses and ruler, using arcs drawn from a point on the line. Here is how.

Take a straight line and call its ends A and B. Choose a point C on the line where you want the 90° angle to stand. This is the point the perpendicular will rise from.

With the point of your compasses at C, open them to any width and draw an arc that crosses the line in two places. Call these two crossings D and E. D and E now sit the same distance on either side of C.

Now open the compasses WIDER than the distance DC — that is, wider than the opening you used in step 2. This wider setting matters: if you kept the original width, the two arcs you are about to draw would meet only back at C itself, and you would get nowhere. So keep the wider setting fixed.

Put the compasses at D with this wider setting and draw an arc above the line. Then, keeping the very same wider setting, put the compasses at E and draw another arc above the line, so that it cuts the first one. Call the point where the two arcs meet F.

Finally, join C to F with your ruler. The line CF stands perpendicular to AB, so the angle between CF and the line — angle FCB — is exactly 90°.

The finished figure looks like this (fig-13): the perpendicular rises from a point near the middle of the line, a small square marks the right angle at its foot, and the two arcs cross each other above it where F is.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (7,-0.1) {$B$};
  \draw (0.29,0.22) -- (0.29,-0.22);
  \draw (0.84,0.34) -- (0.84,-0.34);
  \draw (6.13,0.22) -- (6.13,-0.22);
  \draw (6.60,0.34) -- (6.60,-0.34);
  \draw (3.8,0) -- (3.8,3.2);
  \draw (4.12,0) -- (4.12,0.32) -- (3.8,0.32);
  \draw (2.9,2.55) -- (4.7,3.05);
  \draw (2.9,3.05) -- (4.7,2.55);
\end{tikzpicture}
```

WORKED_EXAMPLE

Let us construct a 90° angle step by step, following the procedure exactly.

1. Draw a straight line AB and mark a point C on it where you want the right angle.
2. Put the compasses at C and draw an arc that cuts the line at two points. Label them D and E.
3. Open the compasses wider than DC — the width from step 2 will not work. With the compasses at D at this wider setting, draw an arc above the line.
4. Keeping the same wider setting, put the compasses at E and draw another arc that cuts the first one at F.
5. Join C to F with a ruler.
6. Since CF is perpendicular to AB, angle FCB = 90°.

The square mark at the foot of CF shows the right angle, and the two short strokes crossing near the top of the perpendicular show where F was found by the two arcs.

1. Draw a straight line AB and mark a point C on it. Using compasses and ruler, construct a perpendicular from C so that angle FCB = 90°. What must you do differently with the compasses between step 2 and step 3, and why?

answer: You must open the compasses WIDER than the distance DC before drawing the arcs from D and E. The original width would not work because arcs of that width from D and E would meet only back at C itself, so they would never cross above the line to give F.

2. In the construction of angle 90°, you draw an arc from point D and an arc from point E that cut each other at F. Why must D and E lie equal distances from C before you do this?

answer: Because the arc drawn from C with the compasses at a fixed opening cuts the line at two points D and E that are each the same distance from C (that radius). With D and E symmetric about C, the two wider arcs drawn from them meet above C at F, so that CF rises straight up from the middle — perpendicular to the line.

Objective 5

Constructing an Angle of 60°

> EXPLANATION
> A 60° angle is one of the most useful angles to construct because, once you can build it, you can build several others from it. All you need is your compass and a straight edge — no measuring of the angle itself. The whole trick is that the two arcs you draw have equal radii, which is what locks the angle at 60°.
>
> Here is the idea in three steps. You will swing two arcs of the same radius, and the point where the second arc meets the first gives you the direction of the arm that makes the 60° angle.

> WORKED_EXAMPLE
> Construct an angle of 60° at point B on a line, following along with fig-14.
>
> **Step 1 — the base line and the first arc.** Draw a line segment BC, with B at the left. Put the point of your compass on B and open it to any convenient radius — say, about half the length of BC. With the centre at B, draw an arc that crosses BC. Mark where that arc cuts BC as the point X.
>
> **Step 2 — the second arc.** Keep the compass at exactly the same radius (do not change the opening). Put the point of the compass on X, and draw an arc. Where this arc cuts your first arc, mark the point A.
>
> **Step 3 — join AB.** Use your straight edge to draw the line from B to A.
>
> The angle ABX — the angle between BA and BC at point B — is 60°.
>
> Why it works: both arcs came from the same radius, so the distance BX equals the distance AX (both are the radius), and BX also equals BA. All three sides are equal, which makes triangle ABX an equilateral triangle, and every angle of an equilateral triangle is 60°.

> VISUAL
> ```
> \begin{tikzpicture}[scale=0.55,line width=0.5pt]
>  \draw (0,0) -- (7,0);
>  \draw (0,0.22) -- (0,-0.22);
>  \node[below left] at (0,-0.05) {$B$};
>  \node[above right] at (7,0) {$C$};
>  \draw (5.17,0.22) -- (5.17,-0.22);
>  \node[below] at (5.17,-0.25) {$X$};
>  \draw (0,0) ++(0:5.17) arc (0:74:5.17);
>  \draw (5.17,0) ++(103:5.17) arc (103:140:5.17);
>  \draw (0,0) -- (2.59,4.48);
> \end{tikzpicture}
> ```

1. Construct an angle of 60° at point P on a horizontal line PQ, using only a compass and a straight edge. List the steps you took.

answer: 1. Draw the line segment PQ with P at the left. 2. With centre P at any convenient radius, draw an arc cutting PQ at a point, call it X. 3. Keeping the same radius, with centre X draw an arc to cut the first arc at a point, call it A. 4. Join P to A. The angle APQ is 60°.

2. In the construction of a 60° angle, why must the second arc be drawn with the same radius as the first?

answer: Because the two equal radii make the two sides from the vertex (and the side between the arc points) all equal, giving an equilateral triangle. Every angle of an equilateral triangle is 60°, so using the same radius is what fixes the angle at 60°.

C3 `C2 plus a validation and repair pass over the draft.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Construction

`subtopic`

Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

`term`

Third Term

`week`

6

`lesson_objectives`

• construct parallel lines
• construct perpendicular lines
• bisect a given line segment
• construct angle 90°
• construct angle 60°

`entry_behaviour`

• Students have been taught how to draw lines.

`instructional_materials`

• Mathematical set.

Objective 1 — construct parallel lines

objective_text

construct parallel lines

current_step

1

instructional_events

• present_content
• elicit_performance

student_tasks

• construct parallel lines, following the teacher's board construction and replicating it in their notes

source_figures


```
[
 {
  "caption": "Line $AB$, with a point $P$ marked above it. A set square is drawn below $AB$ with
one edge along the line: the right angle is at the left end of that edge, one leg runs down from
it, and the hypotenuse closes back to a point on $AB$ to the right.",
  "figure_ref": "fig-01"
 },
 {
  "caption": "The same set square on $AB$, with the ruler now drawn as a long strip lying along
the set square's downward edge. The set square's vertical edge and the ruler's right edge are
**layered** — only one line is visible on the page. The right angle marked at the corner is the
set square's own, between the edge lying along $AB$ and the edge rising from it.",
  "figure_ref": "fig-02"
 },
 {
  "caption": "The same arrangement drawn a second time alongside the first — the set square's
edge again **layered** on the ruler's right edge, with no gap between them — and with the ruler
leaning — running from upper left to lower right across $AB$ instead of standing vertical. Only
$A$ is labelled. The slanted ruler is the point of this panel — it shows that the ruler's
inclination does not affect the construction.",
  "figure_ref": "fig-03"
 },
 {
  "caption": "The vertical ruler held fixed, with the set square drawn twice — once with its edge
on $AB$ and once higher up, its edge on a second line parallel to $AB$ — and an arrow between the
two positions pointing up along the ruler.",
  "figure_ref": "fig-04"
 },
 {
  "caption": "The same slide drawn a second time alongside it, with the leaning ruler. Both set
square positions are **layered** on the ruler's right edge, as in fig-03. Here the upper line
carries the point $P$, and the set square's upper position has its edge running through $P$.",
  "figure_ref": "fig-05"
 }
]
```


lesson_content

```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Examples:** construct a line through Point P so that it is parallel to AB.\n\n(a)
Place a set square so that one edge is accuratly along AB\n\n**fig-01.** Line $AB$, with a point
$P$ marked above it. A set square is drawn below $AB$ with\none edge along the line: the right
angle is at the left end of that edge, one leg runs down from\nit, and the hypotenuse closes back
to a point on $AB$ to the right.\n\n[drawing attached to this section as fig-01, marked A, B, P]
\n\n(b) Place a ruler along one of the other edge of the set square. Use the left hand edge if
you\nare right handed.\n\n**fig-02.** The same set square on $AB$, with the ruler now drawn as a
long strip lying along the\nset square's downward edge. The set square's vertical edge and the
ruler's right edge are\n**layered** — only one line is visible on the page. The right angle
marked at the corner is the\nset square's own, between the edge lying along $AB$ and the edge
rising from it.\n\n[drawing attached to this section as fig-02, marked A, B]\n\n**fig-03.** The
same arrangement drawn a second time alongside the first — the set square's edge\nagain
**layered** on the ruler's right edge, with no gap between them — and with the ruler leaning\n—
running from upper left to lower right across $AB$ instead of standing vertical. Only $A$
is\nlabelled. The slanted ruler is the point of this panel — it shows that the ruler's
inclination\ndoes not affect the construction.\n\n[drawing attached to this section as fig-03,
marked A]\n\n(c) Hold the ruler firmly, slide the set square along the ruler towards P. Stop when
the edge that\nwas on AB reaches P. Draw a line along this edge of the set square through P.
\n\n**fig-04.** The vertical ruler held fixed, with the set square drawn twice — once with its
edge on\n$AB$ and once higher up, its edge on a second line parallel to $AB$ — and an arrow
between the two\npositions pointing up along the ruler.\n\n[drawing attached to this section as
fig-04, marked A, B]\n\n**fig-05.** The same slide drawn a second time alongside it, with the
leaning ruler. Both set\nsquare positions are **layered** on the ruler's right edge, as in
fig-03. Here the upper\nline carries the point $P$, and the set square's upper position has its
edge running through $P$.\n\n[drawing attached to this section as fig-05, marked A, P]"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Objective 2 — construct perpendicular lines

objective_text

construct perpendicular lines

current_step

2

instructional_events

• present_content
• elicit_performance

student_tasks

• follow the teacher's board construction and replicate it in their notes

source_figures


```
[
 {
  "caption": "Line $AB$ with a second line $XY$ crossing it, $X$ above and $Y$ below, and a
right-angle mark in the upper right quadrant at the crossing.",
  "figure_ref": "fig-06"
 },
 {
  "caption": "A plain line with a stroke at each end, labelled $A$ on the left and $B$ on the
right. The point $Q$ named in the instruction is not marked on the drawing.",
  "figure_ref": "fig-07"
 },
 {
  "caption": "Line $AB$ with the set square standing on it as a narrow right-angled triangle: a
short base along $AB$, a near-vertical edge rising from the right end of that base, and the
hypotenuse running up from the left.",
  "figure_ref": "fig-08"
 },
 {
  "caption": "Line $AB$ with $XY$ now drawn across it, $X$ above and $Y$ below, and a right-angle
mark in the upper right quadrant. The foot of $XY$ is not labelled $Q$.",
  "figure_ref": "fig-09"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "Two lines are said to be perpendicular to each other when they meet at a right
angle i.e\n$90^\\circ$\n\n**fig-06.** Line $AB$ with a second line $XY$ crossing it, $X$ above
and $Y$ below, and a\nright-angle mark in the upper right quadrant at the crossing.\n\n[drawing
attached to this section as fig-06, marked A, B, X, Y]\n\n1. Draw a line AB mark a point Q on the
line.\n\n**fig-07.** A plain line with a stroke at each end, labelled $A$ on the left and $B$ on
the right.\nThe point $Q$ named in the instruction is not marked on the drawing.\n\n[drawing
attached to this section as fig-07, marked A, B]\n\n2. Place the edge of the set square on line
AB. Let the edge of the ruler meet the hypotenuse of\n   the set square as shown.\n\n**fig-08.**
Line $AB$ with the set square standing on it as a narrow right-angled triangle: a\nshort base
```

```
along $AB$, a near-vertical edge rising from the right end of that base, and the\nhypotenuse
running up from the left.\n\n[drawing attached to this section as fig-08, marked A, B]\n\n3.
Pressing firmly on the ruler, slide the set square down along the hypotenuse side until the\n
third edges touches point Q. Draw a line XY\n\n**fig-09.** Line $AB$ with $XY$ now drawn across
it, $X$ above and $Y$ below, and a right-angle\nmark in the upper right quadrant. The foot of
$XY$ is not labelled $Q$.\n\n[drawing attached to this section as fig-09, marked A, B, X, Y]"
 }
]
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Objective 3 — bisect a given line segment

objective_text

bisect a given line segment

current_step

3

instructional_events

• present_content
• elicit_performance

student_tasks

• construct line segment, following the teacher's board construction and replicating it in their notes

source_figures


```
[
 {
  "caption": "The line segment $AB$, a stroke at each end, $A$ on the left and $B$ on the
right.",
  "figure_ref": "fig-10"
 },
 {
  "caption": "The segment $AB$ with arc pairs crossing above and below it, the upper crossing
labelled $P$ and the lower one $Q$. Each crossing is drawn as a small X where an arc from $A$
meets an arc from $B$.",
  "figure_ref": "fig-11"
 },
 {
  "caption": "The same arcs with $P$ joined to $Q$, the joining line crossing $AB$ at its middle
and carrying a right-angle mark in the upper right quadrant.",
  "figure_ref": "fig-12"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Procedures:**\n\n1. Open a pair of compasses so that the radius is about $\
\frac{3}{4}$ of the length of AB.\n2. To bisect the line segment AB means to divide it into two
equal halves\n\n**fig-10.** The line segment $AB$, a stroke at each end, $A$ on the left and $B$
on the right.\n\n[drawing attached to this section as fig-10, marked A, B]\n\n3. Place the sharp
point of the compass on A, Draw two arcs, one above and the other below the\n   midle of AB.
\n\n**fig-11.** The segment $AB$ with arc pairs crossing above and below it, the upper
crossing\nlabelled $P$ and the lower one $Q$. Each crossing is drawn as a small X where an arc
from $A$ meets\nan arc from $B$.\n\n[drawing attached to this section as fig-11, marked A, B, P,
Q]\n\n4. With the same radius, and place the sharp point of the compasses on B Draw two arc so
that they\n   cut the first arc at P and Q\n\n**fig-12.** The same arcs with $P$ joined to $Q$,
the joining line crossing $AB$ at its middle and\ncarrying a right-angle mark in the upper right
quadrant.\n\n[drawing attached to this section as fig-12, marked A, B, P, Q]"
```

```
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Objective 4 — construct angle 90°

`objective_text`

construct angle 90°

`current_step`

4

`instructional_events`

• present_content
• elicit_performance

`student_tasks`

• construct angle 90 degrees, following the teacher's board construction and replicating it in their notes

`source_figures`


```
[
 {
  "caption": "The line $AB$ with a stroke at each end, the two arc cuts $X$ and $Y$ marked either side of the middle, and the perpendicular $OZ$ erected from $O$ between them, with a right-angle mark at its foot and the two arcs crossing at $Z$ near its top.",
  "figure_ref": "fig-13"
 }
]
```


`lesson_content`


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Procedure:**\n\n1. Draw a line segment $AB$ and mark a point $O$ on it where the right angle is to stand.\n2. With centre $O$ and any convenient radius, draw an arc to cut $AB$ at $X$ and $Y$.\n3. With centre $X$ and a radius greater than $XO$, draw an arc above $AB$.\n4. With centre $Y$ and the same radius as in (3), draw an arc to cut the first arc at $Z$.\n5. Join $OZ$. Then $\\angle ZOA = \\angle ZOB = 90^\\circ$.\n\n**fig-13.** The line $AB$ with a stroke at each end, the two arc cuts $X$ and $Y$ marked either\nside of the middle, and the perpendicular $OZ$ erected from $O$ between them, with a right-angle\nmark at its foot and the two arcs crossing at $Z$ near its top.\n\n[drawing attached to this section as fig-13, marked A, B, X, Y, O, Z]"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Objective 5 — construct angle 60°

objective_text

construct angle 60°

current_step

5

instructional_events

• present_content
• elicit_performance

student_tasks

• construct angle 60 degrees, following the teacher's board construction and replicating it in their notes

source_figures


```
[
 {
  "caption": "The line $BC$ with $B$ at the left, a stroke marking $X$ part way along, a long arc swung from $B$ through $X$ and up to the left, a shorter arc swung from $X$ cutting it, and a straight line from $B$ to that crossing. The crossing itself is not labelled.",
  "figure_ref": "fig-14"
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]
```


lesson_content


```
[
 {
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  "excerpt": "**Procedure:**\n\n1. Draw a line segment BC with centre B at any convinent radius. Draw an arc to cut BC at X\n2. with centre X and the same radius as in (i) draw an arc to cut the first arc at A\n3. Join AB.\n\n**fig-14.** The line $BC$ with $B$ at the left, a stroke marking $X$ part way along, a long arc\nswung from $B$ through $X$ and up to the left, a shorter arc swung from $X$ cutting it, and a\nstraight line from $B$ to that crossing. The crossing itself is not labelled.\n\n[drawing attached to this section as fig-14, marked B, C, X]"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Constructing Parallel Lines

EXPLANATION
Two lines are parallel when they stay the same distance apart and never meet, however far you draw them. In this section you will construct a line through a given point P so that it is parallel to a given line AB.

You need only two tools from your mathematical set — a set square and a ruler. No compass is required.

Here is the whole idea, in three steps.

EXPLANATION
Step 1 — Lay the set square along AB. Place the set square so that one of its edges lies accurately along the line AB (fig-01). It does not matter which edge you choose or which way round the set square is turned; the only thing that matters is that a straight edge of the set square lies exactly on AB.

VISUAL
```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \node[left] at (0,0) {$A$};
  \node[above] at (7,0) {$B$};
  \fill (3.4,2.8) circle (1.6pt);
  \node[above] at (3.4,2.8) {$P$};
  \draw (2.7,0) -- (2.7,-1.1) -- (5.2,0);
  \draw (2.7,-0.26) -- (2.96,-0.26) -- (2.96,0);
\end{tikzpicture}
```

EXPLANATION
Step 2 — Put the ruler against another edge of the set square. Place the ruler along one of the set square's other edges, holding it there (fig-02).

VISUAL
```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above] at (8,0) {$B$};
  \draw[rounded corners=3pt] (1.29,3.50) rectangle (1.72,-4.12);
  % The set square's vertical edge lies ALONG the ruler's right edge (x=1.72) — in the
  % source the two are layered, so only one line is visible. Drawn collinear, not offset.
  \draw (1.72,0) -- (1.72,-1.9) -- (5.4,0);
  \draw (1.72,-0.26) -- (1.98,-0.26) -- (1.98,0);
\end{tikzpicture}
```

EXPLANATION
If you are right-handed, rest the ruler against the left-hand edge of the set square; if you are left-handed, use the right-hand edge. The ruler may rest against either the upright edge or the sloping edge — fig-03 shows it leaning against the sloping edge, and the construction still works. What matters is that one edge of the set square stays on AB while the ruler rests against another edge.

VISUAL
```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above right] at (8,0) {$B$};
  % The ruler now lies along the sloping edge, extending past it at both ends. Drawn a
  % hairline clear of that edge so the set square stays visible as a triangle.
  \draw[rounded corners=2pt] (-0.171,-3.033) -- (6.760,0.545) -- (6.967,0.145) -- (0.036,-3.433) -- cycle;
  \draw (1.72,0) -- (1.72,-1.9) -- (5.4,0);
  \draw (1.72,-0.26) -- (1.98,-0.26) -- (1.98,0);
\end{tikzpicture}
```

EXPLANATION
Step 3 — Hold the ruler and slide the set square. Keep the ruler firmly in place with one hand. With the other hand, slide the set square along the ruler towards P. Because the ruler is held still, the set square slides without turning — it keeps pointing exactly the same way, just higher up (fig-04).

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above] at (8,0) {$B$};
  \draw (0.71,2.40) -- (8,2.40);
  \draw[rounded corners=3pt] (0.28,3.65) rectangle (0.71,-3.97);
  % ONE set square drawn in two positions, translated straight up the ruler's right edge
  % (x=0.71). Both are congruent — leg 1.21, width 2.68 — so the two hypotenuses are
  % exactly parallel, which is the visible signature of a slide.
  \draw (0.71,0) -- (0.71,-1.21) -- (3.39,0);
  \draw (0.71,-0.26) -- (0.97,-0.26) -- (0.97,0);
  \draw (0.71,2.40) -- (0.71,1.19) -- (3.39,2.40);
  \draw (0.71,2.14) -- (0.97,2.14) -- (0.97,2.40);
  \draw[->] (2.44,0.1) -- (2.44,1.37);
\end{tikzpicture}
```

EXPLANATION

Stop as soon as the edge that was on AB reaches P (fig-05). Now draw a line along that edge of the set square, through P.

This new line through P is parallel to AB. Sliding along a held ruler moves the set square without turning it, so the edge that passes through P keeps the same direction as the edge that lay on AB — and that is exactly what "parallel" means.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below left] at (0,0) {$A$};
  \node[above right] at (8,0) {$B$};
  \draw (0.35,2.2) -- (8,2.2);
  \fill (6.0,2.2) circle (1.6pt);
  \node[above] at (6.0,2.2) {$P$};
  % The ruler is held still; it spans both positions of the set square, drawn a hairline
  % clear of their sloping edges.
  \draw[rounded corners=2pt] (-0.026,-2.119) -- (7.616,2.247) -- (7.824,1.882) -- (0.182,-2.484) -- cycle;
  \draw (0.6,0) -- (0.6,-1.6) -- (3.4,0);
  \draw (0.6,-0.26) -- (0.86,-0.26) -- (0.86,0);
  % The same set square, translated along the ruler until its edge reaches P.
  \draw (4.45,2.2) -- (4.45,0.6) -- (7.25,2.2);
  \draw (4.45,1.94) -- (4.71,1.94) -- (4.71,2.2);
  \draw[->] (2.6,0.55) -- (4.0,1.35);
\end{tikzpicture}
```

WORKED_EXAMPLE

Worked example — construct a line through P parallel to AB.

1. Draw the line AB and mark a point P above it, as in fig-01.
2. Place one edge of the set square accurately along AB.
3. Hold the ruler firmly against another edge of the set square (use the left-hand edge if you are right-handed).
4. Slide the set square along the ruler towards P, without letting it turn, until the edge that was on AB reaches P.
5. Draw a line along that edge of the set square, through P. This line is parallel to AB (fig-05).

1. On a sheet of paper, draw a line AB and mark a point P above the middle of it. Construct a line through P that is parallel to AB. Write the steps you used.

answer: 1. Lay one edge of the set square accurately along AB. 2. Place the ruler firmly against another edge of the set square (left-hand edge if right-handed). 3. Slide the set square along the held ruler towards P until the edge that was on AB reaches P. 4. Draw a line along that edge through P. This line is parallel to AB.

2. When you slide the set square along the held ruler, the line you draw through P stays parallel to AB. Explain why, in your own words.

answer: Because the ruler is held firmly, the set square slides without turning — it keeps pointing the same way. So the edge that passed through P has the same direction as the edge that lay on AB, and two lines with the same direction are parallel.

Perpendicular Lines

EXPLANATION
Two lines are said to be **perpendicular** to each other when they meet at a right angle, that is, at 90°.

Look at fig-06. Line AB is crossed by line XY, with X above and Y below the crossing. The little square mark drawn in the upper right quadrant at the crossing tells you the two lines meet at a right angle. Whenever you see that square corner, the two lines are perpendicular.

VISUAL
```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (-0.5,0) -- (7.5,0);
  \draw (0,0.25) -- (0,-0.25);
  \draw (5,0.25) -- (5,-0.25);
  \node[below] at (0,-0.3) {$A$};
  \node[below] at (5,-0.3) {$B$};
  \draw (2.5,2.6) -- (2.5,-2.6);
  \node[above] at (2.5,2.6) {$X$};
  \node[below] at (2.5,-2.6) {$Y$};
  \draw (2.82,0) -- (2.82,0.32) -- (2.5,0.32);
\end{tikzpicture}
```

EXPLANATION
You can construct a perpendicular line using a ruler and a set square. Here is how.

Step 1 — Draw the line and mark a point.
Draw a line AB and mark a point Q on it, where you want the perpendicular line to pass through. Fig-07 shows the line AB with A on the left and B on the right.

WORKED_EXAMPLE
Step 2 — Place the set square on the line.
Place one edge of the set square along line AB. Now let the edge of the ruler meet the hypotenuse (the sloping edge) of the set square, as fig-08 shows. The set square stands as a narrow right-angled triangle: its short base lies along AB, its upright edge rises from the right end of that base with a right angle marked there, and its hypotenuse closes back down to the left.

Step 3 — Slide the set square down.
Pressing firmly on the ruler, slide the set square down along the hypotenuse side until its third edge (the upright edge) touches the point Q. Then draw the line XY along that edge, with X above and Y below the crossing, as in fig-09. Mark the right angle in the upper right quadrant to show the lines are perpendicular.

VISUAL
```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below] at (0.15,-0.1) {$A$};
  \node[above] at (4.52,0.1) {$B$};
  \draw (2.1,1.85) -- (2.15,-0.8);
  \node[above] at (2.1,1.85) {$X$};
  \node[below] at (2.15,-0.8) {$Y$};
  \draw (2.44,0) -- (2.44,0.32) -- (2.12,0.32);
\end{tikzpicture}
```

VISUAL
```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (8,0);
  \node[below] at (0.41,-0.05) {$A$};
  \node[above] at (3.93,0.05) {$B$};
  % The upright edge stands square on the base: step 2 and 3 name a hypotenuse and a
  % "third edge", which only a marked right angle lets the reader tell apart.
  \draw (1.54,0) -- (2.25,1.0) -- (2.25,0) -- cycle;
  \draw (1.97,0) -- (1.97,0.28) -- (2.25,0.28);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below] at (0,-0.28) {$A$};
  \node[below] at (7,-0.28) {$B$};
\end{tikzpicture}
```

1. Two lines cross each other and the angle between them is 90°. What do we call these two lines?
answer: Perpendicular lines (they meet at a right angle, 90°).

2. Draw a line AB, mark a point Q on it, and construct a line through Q perpendicular to AB using your ruler and set square.
answer: Your constructed line crosses AB at Q with a right angle, so a small square mark in one quadrant at Q confirms the two lines are perpendicular.

3. In the construction, when you slide the set square down along the hypotenuse, what edge of the set square must touch the point Q before you draw the perpendicular line?
answer: The third edge — the upright edge that rises from the base of the set square (the edge carrying the right angle).

Bisecting a Line Segment

EXPLANATION

To bisect a line segment means to divide it into two equal halves. So when you bisect segment AB, you find the point exactly halfway along it — the midpoint M — where AM equals MB.

Here is how to bisect segment AB with a pair of compasses and a ruler.

Step 1 — open the compasses. Open them so that the radius is about three-quarters of the length of AB. (Any opening larger than half the length of AB will work — the important thing is that the arcs you draw next will be big enough to cross.)

WORKED_EXAMPLE

Take the line segment AB shown in fig-10, with A on the left and B on the right.

Step 2 — draw arcs from A. Put the sharp point of the compasses on A. Draw two arcs — one above the middle of AB and one below it (fig-11).

Step 3 — draw arcs from B. Keep the compasses at the same radius. Put the sharp point on B and draw two more arcs so that they cut the first arcs. The arc from A and the arc from B meet at a point P above the line, and at a point Q below it. Each crossing is marked with a small X in fig-11.

Step 4 — join P to Q. Use your ruler to draw the straight line from P to Q (fig-12).

Step 5 — read off the midpoint. The line PQ crosses AB at M. This M is the midpoint of AB, so AM = MB. Notice too that PQ meets AB at a right angle (90°), marked in fig-12.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (6,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (6,0.22) -- (6,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (6,-0.1) {$B$};
  \draw (0,0) ++(37:4.5) arc (37:59:4.5);
  \draw (6,0) ++(121:4.5) arc (121:143:4.5);
  \draw (0,0) ++(-59:4.5) arc (-59:-37:4.5);
  \draw (6,0) ++(-143:4.5) arc (-143:-121:4.5);
  \draw (3,3.35) -- (3,-3.35);
  \node[above] at (3,3.35) {$P$};
  \node[below] at (3,-3.35) {$Q$};
  \node[below right] at (3,-0.1) {$M$};
  \draw (3.32,0) -- (3.32,0.32) -- (3,0.32);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (6,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (6,0.22) -- (6,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (6,-0.1) {$B$};
  \draw (0,0) ++(37:4.5) arc (37:59:4.5);
  \draw (6,0) ++(121:4.5) arc (121:143:4.5);
  \draw (0,0) ++(-59:4.5) arc (-59:-37:4.5);
  \draw (6,0) ++(-143:4.5) arc (-143:-121:4.5);
  \node[above] at (3,3.5) {$P$};
  \node[below] at (3,-3.5) {$Q$};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (7,-0.1) {$B$};
\end{tikzpicture}
```

1. Explain in your own words what it means to bisect the line segment XY.

answer: To bisect segment XY means to divide it into two equal halves — to find the midpoint M so that XM = MY.

2. Describe the compass and ruler construction that finds the midpoint of a given segment AB. Give the steps in order.

answer: Open the compasses to about three-quarters of the length of AB. Put the point on A and draw an arc above and an arc below the line. Keeping the same radius, put the point on B and draw two arcs that cut the first ones at P (above) and Q (below). Join P to Q with a ruler. Where PQ crosses AB is the midpoint M.

3. After bisecting segment AB, the construction line PQ crosses AB at M. Which two lengths are equal at M, and at what angle does PQ meet AB?

answer: AM = MB, and PQ meets AB at a right angle (90°).

Constructing an Angle of 90°

EXPLANATION

You can construct an angle of exactly 90° using only a ruler and a pair of compasses — no protractor needed. Here is the step-by-step method.

Step 1 — Draw your line and mark the point. Draw a straight line and label its ends A and B. Mark a point C on the line where you want the 90° angle to be built.

Step 2 — Strike an arc that cuts the line twice. Put the point of the compasses at C and draw an arc. The arc crosses the line at two points; label them D and E. The width you used here is the distance DC.

Step 3 — Open the compasses wider, then arc from D. Open the compasses wider than DC. This wider setting matters: if you tried to use the step-2 width, the next two arcs would only meet at C itself, which gives you nothing. With the compasses set wider and placed at D, draw an arc above the line.

Step 4 — Arc from E at the same wider setting. Keeping the compasses at the same wider setting, place them at E and draw another arc above the line so that it cuts the first arc. Label the crossing point F.

Step 5 — Join C to F. Use your ruler to draw the straight line from C up to F.

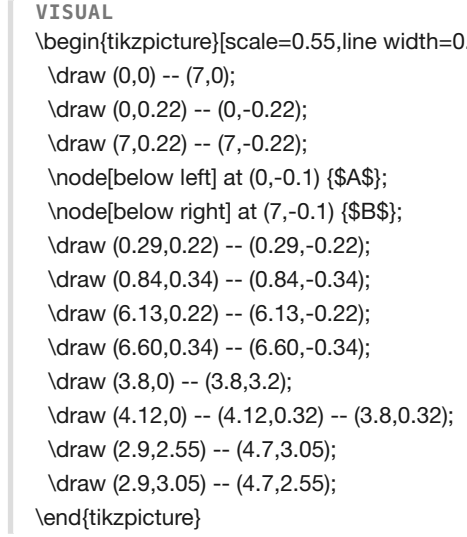
Step 6 — Read the angle. The line CF is perpendicular to AB, so the angle between CF and the line AB — written ∠FCB — is exactly 90°.

WORKED_EXAMPLE

Follow the construction for a line AB with point C marked on it, where the 90° angle is to stand.

1. With the compasses at C, draw an arc that cuts the line at D and E.
2. Open the compasses wider than DC. Place them at D and draw an arc above the line.
3. Keeping the same wider setting, place the compasses at E and draw another arc that cuts the first one at F.
4. Join C to F with a ruler.
5. CF is perpendicular to AB, so ∠FCB = 90°. A right-angle mark at the foot of CF confirms the corner is square.

VISUAL



```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \draw (7,0.22) -- (7,-0.22);
  \node[below left] at (0,-0.1) {$A$};
  \node[below right] at (7,-0.1) {$B$};
  \draw (0.29,0.22) -- (0.29,-0.22);
  \draw (0.84,0.34) -- (0.84,-0.34);
  \draw (6.13,0.22) -- (6.13,-0.22);
  \draw (6.60,0.34) -- (6.60,-0.34);
  \draw (3.8,0) -- (3.8,3.2);
  \draw (4.12,0) -- (4.12,0.32) -- (3.8,0.32);
  \draw (2.9,2.55) -- (4.7,3.05);
  \draw (2.9,3.05) -- (4.7,2.55);
\end{tikzpicture}
```

1. Construct an angle of 90° on a fresh line. Mark your own point C on the line, then use compasses and ruler to build the perpendicular. Check your result: the angle between the raised line and the original line should be exactly 90°.

answer: The raised line CF is perpendicular to the original line AB, so ∠FCB = 90°.

2. In the 90° construction, why must you open the compasses wider than DC in step 3, instead of keeping the width you used when drawing the arc from C?

answer: Arcs drawn at the step-2 width from D and E would only meet at C itself, which gives no new crossing point F. Opening the compasses wider than DC makes the two arcs cut each other at a point F above the line, which is what you join to C.

Constructing a 60° Angle

EXPLANATION

An angle of 60° is one you can draw with your mathematical set using only a pencil, a ruler and a pair of compasses. The secret is to draw arcs that all have the same radius. When you swing an arc from one end of a line and then swing another arc of the same radius from a point on that line, the two arcs cross at a point that lies exactly 60° above the line. Here is why it works: because both arcs share the same radius, the distance from the start point to the crossing, the distance from the start point to the mark on the line, and the distance from that mark to the crossing are all equal. That makes the triangle equilateral, and an equilateral triangle has every angle equal to 60°, so the angle at the start point is exactly 60°.

You will need a ruler, a pencil and a pair of compasses. The steps below give you the whole construction.

WORKED_EXAMPLE

Worked example — construct a 60° angle

Step 1. Draw a straight line and mark the two ends of it. Call the left end B and the right end C, so the line is BC.

Step 2. Put the point of your compasses on B. Open the compasses to any convenient radius — it can be any size you like. Swing an arc from B so that it crosses the line BC. Where the arc meets the line, mark the point X. The arc passes through X and curves upward to the left.

Step 3. Keeping the compasses at the same radius, move the point to X. Swing another arc from X so that it cuts the first arc. Call the point where the two arcs cross A.

Step 4. Join B to A with a straight line using your ruler. The angle ABC, between the line BA and the line BC, is 60°.

Because you kept the same radius for both arcs, the distance from B to X, the distance from X to A, and the distance from B to A are all equal. So triangle ABX is equilateral, and in an equilateral triangle every angle is 60°, making the angle ABC exactly 60°.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) -- (7,0);
  \draw (0,0.22) -- (0,-0.22);
  \node[below left] at (0,-0.05) {$B$};
  \node[above right] at (7,0) {$C$};
  \draw (5.17,0.22) -- (5.17,-0.22);
  \node[below] at (5.17,-0.25) {$X$};
  \draw (0,0) ++(0:5.17) arc (0:74:5.17);
  \draw (5.17,0) ++(103:5.17) arc (103:140:5.17);
  \draw (0,0) -- (2.59,4.48);
\end{tikzpicture}
```

1. Use your mathematical set to construct a 60° angle. Start by drawing a line, marking its two ends as B and C, and follow the compass-arc steps to produce the angle. When you finish, measure the angle you have drawn with your protractor.

answer: Your construction should show an angle of 60°.

2. In the construction of a 60° angle, you swing two arcs. What must be true about their radii for the construction to work, and why does it matter?

answer: The two arcs must have the same radius, because the equal spacing they produce is what makes the angle exactly 60°.

Result — per rubric dimension

Scores are 1–5. Every figure is the mean of 3 draws rated by 2 judge pass(es) — 6 judgments per cell.

Rubric	C0	C1	C2	C3	C1 – C0	C2 – C1	C3 – C2
Objective and curriculum alignment	5.00	5.00	5.00	5.00	+0.00	+0.00	+0.00
Taught-lesson alignment	3.50	4.00	3.50	3.67	**+0.50**	**-0.50**	**+0.17**
Scope and conceptual correctness	4.67	5.00	4.50	5.00	**+0.33**	**-0.50**	**+0.50**
Student-facing clarity and usefulness	4.67	4.50	3.83	4.00	**-0.17**	**-0.67**	**+0.17**

Why the judge scored it that way

Every rationale the judge wrote for this lesson, deduplicated across draws and passes. These are its words, not a summary.

Objective and curriculum alignment — C0 5.00, C1 5.00, C2 5.00, C3 5.00

- `C0` The revision pack maps one-to-one onto every stated objective and covers the full JSS1 construction curriculum target for this week without omitting or substituting any item.
- `C0` The revision pack covers every stated objective in the lesson plan, in the correct order, with appropriate depth for JSS1. There are no missing objectives and no extraneous topics that fall outside the curriculum target.
- `C0` The revision pack maps one-to-one onto every stated lesson objective and covers each with explanation, worked example, and practice questions. There is no gap between the objectives and the content provided.
- `C0` The revision pack maps one-to-one onto every stated lesson objective and covers all five construction tasks required by the curriculum target. Nothing is omitted.
- `C0` Every stated lesson objective is addressed in a dedicated section with explanation, worked examples, and practice questions. There are no missing objectives and no objectives from outside this lesson's scope.
- `C0` The revision pack covers every single objective from the lesson plan, one dedicated section per objective, with no objective omitted or added. Alignment with the JSS1 Third Term curriculum target for construction is complete.
- `C1` The revision pack covers every stated lesson objective completely and in the correct curriculum order for JSS1 Third Term construction work. No objective is omitted or diluted.
- `C1` The revision pack covers every stated lesson objective completely and in the correct curriculum context, with no objectives omitted or added beyond scope.
- `C1` Every one of the five stated lesson objectives has its own dedicated section with explanation, worked example, and practice questions. There are no gaps and no off-topic material.
- `C1` The revision pack covers every one of the five stated lesson objectives with dedicated explanatory text, a worked example, and practice questions, making the alignment complete and unambiguous.
- `C1` The revision pack comprehensively addresses every stated lesson objective in the correct curriculum context, with no objective omitted or substituted. Alignment is complete.
- `C1` The revision pack covers every single one of the five lesson objectives completely and in order, with no gaps and no objectives from outside the lesson. Alignment with the stated curriculum target is perfect.
- `C2` The revision pack maps one-to-one onto every stated objective without omission or addition, giving complete coverage of the planned curriculum target.
- `C2` Every stated objective in the lesson plan has a dedicated section in the revision pack, and the content aligns with the JSS1 curriculum expectation for geometric construction. Nothing important is omitted.
- `C2` The revision pack covers every one of the five stated objectives in the correct curriculum context for JSS1 Term 3, with no objectives omitted and none added from outside the lesson scope.
- `C2` The revision pack maps exactly onto every stated lesson objective without omitting any. Coverage is thorough and nothing outside the stated curriculum target is introduced.
- `C2` The revision pack maps one-to-one onto every stated lesson objective, covering each construction in its own dedicated section with explanation, worked example, and practice. There are no gaps and no additions that fall outside the curriculum target.

- `C2` Every stated lesson objective has a corresponding section in the revision pack, covered at the appropriate JSS1 depth. The alignment is complete and faithful.
- `C3` Every one of the five lesson objectives is addressed thoroughly and in order. There are no missing objectives and no extraneous topics that would distract from the curriculum target.
- `C3` The revision pack maps one-to-one onto every stated lesson objective and aligns with the JSS1 curriculum scope for geometric construction. Nothing is omitted and nothing extraneous is added.
- `C3` The pack covers every one of the five stated lesson objectives fully and in the correct curriculum context. There are no gaps and no extraneous objectives introduced.
- `C3` The revision pack maps one-to-one onto every stated lesson objective and the curriculum subtopic, with no objective missing or substituted. Coverage is complete and well-matched to the JSS1 Third Term specification.
- `C3` The revision pack covers every single objective stated in the lesson plan with dedicated, appropriately detailed sections. There are no gaps and no out-of-scope objectives included.
- `C3` Every stated objective has a full section with explanation, worked example, and practice questions. There are no gaps and no objectives addressed that fall outside the lesson.

Taught-lesson alignment — C0 3.50, C1 4.00, C2 3.50, C3 3.67

- `C0` The pack faithfully mirrors the lesson's sequence, tools, and methods; the only minor gap is that the lesson describes students replicating constructions in their notes as a hands-on activity, and the pack's practice tasks are somewhat more wordy/explanatory than purely practical, but the depth and emphasis remain well aligned.
- `C0` The pack broadly follows the taught sequence and methods, and the step-by-step format matches classroom replication activities. However, the lesson plan's distinction between drawing perpendicular lines (using set square and ruler as a primary tool) and constructing 90° by compass is blurred in the pack, which uses compass arcs for both, slightly misrepresenting the lesson's emphasis and tool usage for the perpendicular section.
- `C0` The pack faithfully mirrors the methods, sequence, and tools the teacher used in class. A small gap is that the lesson places bisection before the dedicated perpendicular construction, yet the pack's perpendicular section already introduces bisection concepts — a minor ordering overlap that does not seriously distort alignment.
- `C0` The pack reproduces the lesson's sequence, tools, and methods faithfully. The only slight gap is that the pack uses formal written steps rather than the board-and-replicate activity format, but this is appropriate for a take-home revision context and the depth and emphasis match the taught lesson well.
- `C0` The section order and topic coverage match the lesson plan well, but the method for constructing perpendicular lines diverges from the teacher's set-square-and-ruler approach, introducing a compass construction the teacher did not demonstrate. This is a notable misalignment in method, though the other four sections align reasonably with the taught activities.
- `C0` The sequence and topics match the taught lesson, and most methods align, but the perpendicular-lines section replaces the teacher's set-square-and-ruler method with a compass-and-arcs construction, which is a mismatch in method and emphasis for that particular objective. The lesson emphasises student replication of board work step by step, and the pack captures that style reasonably well overall, earning a mid-range score.
- `C1` The pack faithfully reflects the tools, methods, and sequence of the taught lesson. A minor limitation is that the lesson emphasised students replicating constructions on paper (hands-on copying from the board), but the pack's explanatory prose is more text-heavy than activity-driven; it still guides students through the same constructions, so alignment is strong but not perfect.
- `C1` The pack closely mirrors the lesson's sequence, methods, and instruments. A minor gap is that the lesson used a set square and ruler for parallel and perpendicular lines specifically, which the pack does represent, but the pack's depth and written explanation style go somewhat beyond the replication-and-practice format of the classroom lesson; nonetheless, alignment is strong overall.
- `C1` Sequence, tools, and methods all match the taught lesson closely. The slight reduction from 5 is because the pack cannot fully capture the teacher's oral explanation and live demonstration dynamic, and the depth of the 'why it works' explanations goes somewhat beyond what a typical JSS1 board lesson would emphasise.
- `C1` The pack faithfully reproduces the sequence and methods of the taught lesson, but its format (prose explanation plus written practice) differs from the lesson's primary activity mode of teacher demonstration followed by student replication, so it is strong but not a perfect reflection of the lesson's activity style.
- `C1` The pack faithfully mirrors the lesson's sequence, methods, and tools, and the additional explanatory depth is appropriate for unsupported home study. A minor gap is that the lesson emphasised students replicating the teacher's exact board steps, whereas the pack adds conceptual justification not explicitly part of the taught lesson.

- `C1` The pack faithfully mirrors the lesson's sequence, tools, and methods. The only minor gap is that the lesson is a largely teacher-demonstration-then-student-replication class activity, and the pack, while providing worked examples and practice, cannot fully reproduce the live, stepwise board-demonstration aspect; but this is inherent to take-home material rather than a flaw in alignment.
- `C2` The pack faithfully follows the teacher's method and order for all five topics. The one small gap is that the lesson emphasises students physically replicating the constructions on their own paper under teacher guidance, whereas the pack's practice tasks are brief and some are verbal/written rather than fully hands-on; depth of guided practice is slightly lighter than the lesson implies.
- `C2` The sequence and methods align well with the taught lesson, but the pack's depth of written explanation and geometric reasoning (e.g. equilateral triangle proof for 60°, sliding invariance argument for parallels) exceeds what JSS1 students would have encountered in a demonstration-and-replicate lesson, making alignment strong but not perfect.
- `C2` The pack faithfully reflects the lesson's sequence, tool choices, and activity style (observe steps, then practise). A very minor gap is that the lesson mentions the 'set square and ruler' method for perpendicular lines while the compass-arc method is the dominant presentation for 90°; the pack keeps these separate and clear, so alignment remains strong.
- `C2` The pack follows the correct sequence and uses the correct instruments for each construction, aligning well with the lesson plan. However, the lesson's core activity was students replicating teacher constructions step by step; the pack is far more verbally descriptive and explanatory than the lesson's practical, watch-and-copy approach, representing a difference in emphasis and depth of exposition.
- `C2` The pack follows the lesson's order and uses the correct tools for most objectives, but the perpendicular-lines section diverges from the taught method: the lesson shows a compass-arc approach from a point, while the pack teaches a set-square slide, reducing alignment for that key objective. All other sections match the lesson's method and expected depth reasonably well.
- `C2` The pack covers the correct tools and methods in the correct sequence, but the lesson's emphasis was on students replicating board constructions step by step, whereas the pack leans on lengthy prose and absent figures. The practical, activity-centred character of the lesson is only partially reflected.
- `C3` The pack closely follows the lesson sequence, uses the same tools and methods the teacher demonstrated, and targets the same constructions. The slight addition of conceptual reasoning (e.g., equilateral-triangle argument for 60°) goes a little beyond what the lesson notes indicate was taught, but this is a minor and beneficial extension rather than a mismatch.
- `C3` The pack faithfully reproduces the lesson's sequence, tools, and step-by-step method. The only slight gap is that the lesson's emphasis on students physically constructing alongside the teacher is partially replaced by descriptive text, but the pack still strongly reflects what was taught and how it was taught.
- `C3` The pack matches the lesson's sequence, tools, and methods well, but its depth of explanation — extended prose justifications, multiple worked-figure references, handedness notes — exceeds the practise-and-replicate emphasis of the taught lesson. It aligns with objectives but somewhat overreaches the expected lesson depth for JSS1.
- `C3` The pack faithfully reflects the taught lesson's sequence, tools, and construction-by-replication approach. A minor shortfall is that the lesson emphasised the teacher drawing on the board while students copy simultaneously, a live-practice depth the take-home pack can only partially replicate, but the procedural detail and figures compensate well.
- `C3` The pack faithfully follows the taught sequence and uses the correct tools for each construction, reflecting the lesson well. The extra conceptual explanation of why the method works is a minor expansion beyond what the lesson plan describes but does not misrepresent the lesson.
- `C3` The pack follows the same order and uses the same tools as the taught lesson, but it adds explanatory depth (geometric reasoning, 'why it works' paragraphs) that goes beyond what the lesson plan's teacher-demonstration-and-replicate model would have established in class, so alignment is good but not perfect.

Scope and conceptual correctness — C0 4.67, C1 5.00, C2 4.50, C3 5.00

- `C0` All constructions are described with correct geometry; no errors or misconceptions were found, and the pack stays within the JSS1 construction boundary without introducing out-of-scope content.
- `C0` The pack is conceptually and mathematically correct throughout, staying well within JSS1 scope. Minor informal phrasing in one explanation does not introduce an actual error, and no content strays beyond the lesson boundary.
- `C0` Every construction method is mathematically sound, the 'why it works' explanations are correct, and the content stays strictly within the expected JSS1 boundary. No errors or misconceptions were found.
- `C0` All five constructions are explained without any mathematical or conceptual errors, and the explanations stay within the JSS1 scope — no content is introduced that goes beyond the lesson boundary.

- `C0` No mathematical or conceptual errors are present anywhere in the pack; all constructions and their justifications are correct. The only concern is mild scope expansion in the perpendicular section, which uses a compass method rather than the set-square method the lesson intended.
- `C0` Every construction method, geometric claim, and numerical answer in the pack is mathematically correct, and nothing exceeds the JSS1 construction syllabus. There are no conceptual or procedural errors to penalise.
- `C1` All five constructions are described with correct steps and correct mathematical justifications. No errors, misconceptions, or out-of-scope content were found; the material stays firmly within JSS1 construction boundaries.
- `C1` All constructions are described with correct mathematical reasoning, no conceptual errors are present, and the content stays appropriately within the JSS1 construction boundary without venturing into more advanced geometry.
- `C1` All five constructions are described with correct procedure and sound mathematical reasoning. No errors, misconceptions, or out-of-scope content were found.
- `C1` All five constructions are described using correct geometric reasoning with no mathematical errors; the scope stays within JSS1 construction content without introducing extraneous or advanced material.
- `C1` All five constructions are presented with correct procedures and sound mathematical reasoning throughout. No errors in scope, method, or concept were identified.
- `C1` Every construction method, every geometric justification, and every numerical answer in the pack is mathematically correct. The pack stays strictly within JSS1 construction scope — no protractor use, no trigonometry, no content beyond what was taught.
- `C2` Every construction procedure is geometrically correct, the underlying reasoning is accurate for JSS1 level, and no topic wanders outside the lesson boundary.
- `C2` All five constructions are described without mathematical errors, the geometric reasoning given for each is sound, and the pack stays within JSS1 construction scope without introducing out-of-syllabus content.
- `C2` Every construction procedure and all mathematical statements in the pack are correct; there are no conceptual errors, over-simplifications that introduce falsehoods, or out-of-scope content such as trigonometry or proof beyond what JSS1 requires.
- `C2` All five constructions are conceptually correct and the mathematical reasoning given is accurate throughout. The scope stays within the lesson boundary; no extraneous topics are introduced. One minor point: the pack's description of the 60° construction places the second arc from X rather than from B, which is a valid but slightly non-standard presentation compared to the classical method, though it is internally consistent and correct.
- `C2` The mathematical content throughout is accurate and the conceptual explanations are sound. The only minor concern is that the perpendicular-lines section teaches a set-square method rather than a compass method, which is a pedagogical mismatch with the lesson but not itself a mathematical error; all constructions described are geometrically correct.
- `C2` All five constructions are mathematically correct and no content falls outside JSS1 scope. One minor issue is that the parallel-lines method (set square slide) is described at slightly greater length than needed for JSS1, but there are no conceptual errors.
- `C3` All five constructions are described with correct procedures and accurate mathematical justifications. No errors, misconceptions, or out-of-scope content were found.
- `C3` All constructions are mathematically correct, the reasoning given for each method is sound, and the pack stays strictly within the JSS1 construction boundary without introducing errors or out-of-scope content.
- `C3` All five constructions are described with correct procedures and correct mathematical reasoning. No errors, misconceptions, or out-of-scope content (e.g., trigonometry, proofs) were introduced.
- `C3` Every construction procedure, geometric justification, and mathematical claim in the pack is correct and stays squarely within the JSS1 construction boundary. There are no errors, over-reaches into higher content, or conceptual omissions.
- `C3` Every construction is mathematically correct, the reasoning given is sound, and nothing strays outside the JSS1 construction curriculum. No errors or misconceptions were found.
- `C3` All five constructions are mathematically correct, the tools specified match the operations (set square/ruler for parallel and perpendicular; compasses for bisection and angle constructions), and nothing in the pack falls outside the JSS1 construction boundary.

Student-facing clarity and usefulness — C0 4.67, C1 4.50, C2 3.83, C3 4.00

- `C0` The pack is well-structured, uses age-appropriate language, and provides enough scaffolding for independent home use; some sections are fairly text-heavy which may challenge weaker readers, but the logical layout and answer key largely compensate.

- **C0** The pack is exceptionally well-structured for independent home use: clear language, concrete steps, worked examples, and self-check answers give a JSS1 learner everything needed to revise and practice without teacher support.
- **C0** The pack is well-structured, clearly written, and genuinely useful for independent revision. The absence of diagrams or figures is a notable limitation for a geometry construction topic, which lowers the score slightly from a maximum.
- **C0** The pack is exceptionally well structured for independent home use: clear language, logical progression within each section, concrete worked examples, and complete answer keys all make it highly useful for a JSS1 learner revising without a teacher present.
- **C0** The pack is exceptionally well organised for independent use: clear steps, relatable language, worked examples, and detailed answer keys all make it highly usable for a JSS1 student working alone at home.
- **C0** The pack is exceptionally well structured for independent home use: friendly language, explicit steps, worked examples, practice questions, and complete answers with explanations all appear in every section. A JSS1 learner could realistically work through this without additional help.
- **C1** The pack is well-structured and language is appropriate for JSS1 students working alone at home. The slight drawback is that the prose explanations are quite lengthy and some sections (particularly the 90° construction) may feel dense for weaker students; a few diagrams would substantially improve usability, but the text alone is still functional.
- **C1** The pack is well-structured, uses accessible language, and includes both practice questions and answers, making it genuinely useful for home revision; the main limitation is the volume of prose in some sections, which could be daunting for lower-ability JSS1 students working independently.
- **C1** The pack is well-organised, practically actionable, and self-contained with answers — strong take-home material. The conceptual 'why' passages are a little dense for lower-ability JSS1 learners but do not undermine the overall usability for the target audience.
- **C1** The language is accessible, the structure is logical and consistent, the worked examples and answer key are thorough, and the relatable analogies make this pack highly suitable for independent home revision by a JSS1 learner.
- **C1** The pack is exceptionally well-structured for independent home use: clear step-by-step instructions, relatable real-world references, practice questions with full solutions, and accessible explanations all combine to make it highly useful for a JSS1 student working alone.
- **C1** The pack is exceptionally well-structured for independent home use: definitions, step-by-step methods, worked examples, graded practice questions, and a complete answer key are all present and written in accessible language appropriate for JSS1 learners.
- **C2** The step-by-step writing and answer key make the pack largely self-sufficient for home use, but the absence of real drawn figures (replaced by text descriptions of figures) is a notable limitation for a construction topic where visual guidance is essential for JSS1 learners.
- **C2** The pack is well-structured with clear steps, practice questions, and full answers, making it genuinely useful for home revision; however, the language in places is above typical JSS1 reading level and the frequent references to figures that may not reproduce clearly could hinder independent use.
- **C2** The language, structure, and answer key make the pack very useful for self-study, but the absence of actual figures (replaced by textual figure descriptions) is a practical limitation for a take-home pack that relies heavily on visual demonstration of construction steps.
- **C2** The pack is well-structured and the language is largely appropriate for JSS1 learners, with clear steps and self-check answers. Its main weakness for home use is that the actual figures are absent and replaced by text descriptions, which significantly limits usability for construction topics that are inherently visual and hand-drawn.
- **C2** The pack is well-structured, logically sequenced, and includes self-check answers, making it genuinely useful for home revision. The absence of actual drawn figures (only text descriptions are present) is a notable limitation for a construction topic where visual reference is critical, preventing a top score.
- **C2** The pack has a clear structure and correct language level, but the complete absence of actual figures — which are central to understanding geometric constructions — is a critical gap for a student revising at home without a teacher present.
- **C3** The pack is well-organised, logically sequenced, and largely accessible, with self-check answers adding real value for independent revision. The absence of actual drawn figures (replaced by text descriptions) and occasionally dense prose reduce its standalone clarity for a JSS1 student working alone at home, preventing a full 5.
- **C3** The pack is well-structured for independent home use: clear language, logical sequencing, worked examples, and self-check answers. It is slightly dense in places (e.g., the parallel-lines explanation with multiple figure variants), which may challenge weaker JSS1 readers, but overall it is highly usable as take-home revision material.

- C3 The pack is well-organised, clearly written, and self-contained with model answers, making it genuinely useful for independent home revision. Its main limitation is that all figures are described textually rather than drawn, which could hinder a JSS1 learner trying to follow the geometric steps without a visual reference.
- C3 The language, structure, and self-marking answers make the pack well-suited for independent revision, but the absence of actual diagrams — replaced only by text descriptions of what figures would show — is a real limitation for a topic where visual step-by-step diagrams are essential for JSS1 learners doing constructions at home.
- C3 The pack is well-structured and largely accessible for JSS1 learners, with clear step-by-step instructions and self-check answers. The reliance on described-rather-than-drawn figures is a usability limitation that prevents a top score, but the overall material is highly useful as take-home review.
- C3 The language and step-by-step structure are well-suited to independent JSS1 use, and the answer key supports self-marking; however, relying on described figures rather than actual drawn figures is a real obstacle for a construction topic where visual guidance is essential.